Faculty of Computers & Information

Smart Surveillance and Safety System for Abnormal and Suspicious Activities

UNDER SUPERVISION

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ABSTRACT

Traditional surveillance systems depend heavily on continuous human monitoring and passive video recording, limiting their ability to perform automated threat detection, real-time behavioral analysis, and rapid response to critical events. These limitations can reduce operational efficiency and increase the risk of delayed or missed incident detection.

This project presents **Camguard**, an intelligent surveillance and safety system that integrates artificial intelligence and environmental sensors to monitor environments, detect abnormal or dangerous events, and generate instant alerts for rapid and effective response while minimizing the need for constant human supervision.

The proposed framework consists of a centralized backend system that coordinate communication between multiple system components through secure APIs. An AI analysis service processes live camera streams to detect security and safety threats such as violence, theft, vandalism, fire, smoke and gas leaks, explosions, weapon presence, restricted-area intrusion, camera tampering, and abnormal human behavior. In parallel, an IoT layer collects environmental sensor data and contributes to event verification using sensor fusion techniques to improve detection accuracy and reduce false alarms.

The system also includes a web-based administrative platform for managing users, cameras, sensors, automation rules, alerts, and AI scheduling operations, in addition to a mobile application that enables authorized users to monitor live streams, receive alerts, review recorded events, and access incident snapshots remotely.

The proposed architecture emphasizes scalability, reliability, modularity, and security through centralized orchestration, encrypted communication, and role-based access control. By combining AI-driven video analytics with IoT-based environmental monitoring, CamGuard provides a robust and adaptive smart surveillance solution suitable for residential, commercial, industrial, healthcare, and public-sector applications.

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SYMBOLS & ABBREVIATIONS

AI: Artificial Intelligence
API: Application Programming Interface
APA: American Psychological Association
CNN: Convolutional Neural Network
CPU: Central Processing Unit
CCTV: Closed-Circuit Television
CV: Computer Vision
DB: Database
DL: Deep Learning
DVR: Digital Video Recorder
FPS: Frames Per Second
GAN: Generative Adversarial Network
GPU: Graphics Processing Unit
GRU: Gated Recurrent Unit
HTTP: HyperText Transfer Protocol
HTTPS: HyperText Transfer Protocol Secure
ID: Identifier
IEEE: Institute of Electrical and Electronics Engineers
IoT: Internet of Things
IP: Internet Protocol
JSON: JavaScript Object Notation
LAN: Local Area Network
LSTM: Long Short-Term Memory
ML: Machine Learning
MQTT: Message Queuing Telemetry Transport
NVR: Network Video Recorder
OCR: Optical Character Recognition
PTZ: Pan / Tilt / Zoom
RBAC: Role-Based Access Control
REST: Representational State Transfer
RTSP: Real-Time Streaming Protocol
SMS: Short Message Service
TLS: Transport Layer Security
UI: User Interface
UX: User Experience
WAN: Wide Area Network
YOLO: You Only Look Once

CHAPTER 1

SYSTEM OVERVIEW

1. INTRODUCTION

Surveillance is the process of observing and monitoring places, people, or activities to detect risks and record events. It is used in many areas such as homes, companies, factories, hospitals, and public facilities. Surveillance is important because it helps protect people and property, and supports safe daily operations. Modern places may face different risks like fires, unauthorized entry, theft, workplace violence, and dangerous crowding. A good surveillance system can detect problems early, improve awareness of what is happening, and help staff make quick decisions during emergencies. It also helps with safety rules, keeping records of incidents, monitoring operations, and supporting investigations after an event. New technologies can improve surveillance systems and make them smarter. Instead of only recording video, system can understand what is happening and react faster *[1]*.

1.1. Artificial Intelligence Overview

Artificial Intelligence (AI) enables computational systems to perform tasks that typically require human intelligence, including data-driven learning, decision-making, and visual understanding. A key subfield, Machine Learning **(**ML**)** identifies patterns in data to facilitate classification or prediction, with traditional approaches relying on manual feature extraction and algorithms such as Support Vector Machines **(**SVMs**)** which optimize class separation, and K-Nearest Neighbors **(**KNN**)** which classifies instances based on proximity in feature space. Deep Learning **(**DL**)** is a subset of ML, employs multi-layer neural networks to automatically extract hierarchical features from large-scale datasets, with commonly used architectures including Convolutional Neural Networks **(**CNNs**)** for spatial data, Recurrent Neural Networks **(**RNNs**)** for sequential data, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks for modeling long-term dependencies, Transformers for attention-based sequence modeling, Autoencoders for feature extraction and dimensionality reduction, and Generative Adversarial Networks (GANs) for synthetic data generation. The field of Computer Vision focuses on the computational interpretation of images and video, supporting tasks such as object classification, detection, segmentation, facial recognition, tracking, and activity analysis *[2]*.

Table 1.1: AI Applications

Application Field	Description
Healthcare	Supports medical diagnosis from images (X-ray/MRI), patient monitoring and disease prediction.
Finance and Banking	Detects fraud, supports credit scoring, risk analysis, and automated customer service.
Education	Enables adaptive learning, automated grading support, and analytics for student performance improvement.
Surveillance and Security	Enables threat detection, face recognition, intrusion detection, abnormal behaviour analysis, and real-time alerting.

1.2. Internet of Things (IoT) and Embedded Systems Overview

The Internet of Things (IoT) is a network of connected devices and sensors that collect data from the environment and send it to a system for processing, such as measuring smoke, gas leaks, temperature, motion, and vibration. IoT enables real-time monitoring, predictive analytics, and automated decision-making across diverse applications, ranging from smart homes to industrial systems. Embedded systems are small, purpose-built computers integrated within devices to perform specific tasks efficiently, reliably, and with minimal power consumption. These systems serve as the computational backbone of IoT devices, executing local data processing, controlling actuators, and ensuring seamless interaction with the network. Together, IoT and embedded systems form a synergistic framework that facilitates intelligent, responsive, and connected environments, enhancing operational efficiency, safety, and overall user experience.

1.2.1 Components of IoT

IoT systems consist of several key components, including sensors and actuators for data collection and physical interaction, connectivity modules to enable communication, data processing units to analyze information, and user interfaces for monitoring and control. These components work together to create a seamless ecosystem for IoT functionality.

1.2.2 IoT Applications in Various Fields

This table highlights various IoT applications across industries: from healthcare monitoring and agricultural efficiency to smart city management, industrial automation, and surveillance and security, showcasing the transformative role of IoT in improving operations and daily life.

Table 1.2 : IOT Applications

Application Field	Description
Healthcare	Uses IoT devices to continuously monitor patient health, improve diagnosis, and enable remote care.
Agriculture	Enhances agricultural efficiency through smart irrigation, crop monitoring, and resource optimization.
Smart City Management	Supports intelligent traffic control, waste management, energy optimization, and urban planning.
Surveillance and Security	Enables real-time monitoring, threat detection, and enhanced security using connected sensors and cameras.
Industrial Automation	Improves manufacturing processes using real-time monitoring, predictive maintenance, and automation.

1.3. Problem Statement

Traditional surveillance systems primarily rely on human monitoring, which creates significant limitations in accurately detecting threats in real-time. This dependence results in slow detection and inadequate responses to critical incidents, including intrusions, violent activities, fires, and gas leaks. As environments such as residential, commercial, and public-sector facilities grow in complexity and size, the inefficiency and high error rates of manual surveillance methods become increasingly problematic, leading to escalating costs and potential safety hazards *[3]*.

One of the primary root causes of these issues is the heavy reliance on human oversight. Many commercial and industrial sites still utilize passive CCTV systems alongside human monitors, which can delay threat detection and reduce responsiveness. Furthermore, existing CCTV systems lack the automation necessary to identify high-risk events, such as fires and armed robberies, further compromising safety in vulnerable settings *[4]*.

Additionally, current surveillance systems often fail to understand and classify events accurately, resulting in a high rate of false alarms. This insufficiency contributes to delayed emergency responses, as the sluggish detection of incidents, paired with poor coordination among emergency teams, can lead to increased risks of loss of life and property damage. Furthermore, the failure to integrate information about multiple hazards such as theft and violence makes it challenging to manage simultaneous threats effectively *[5]*.

Supporting these concerns, alarming statistics highlight the urgent need for improvement. In 2022, U.S. fire departments responded to over one million fires, with structure fires causing the majority of property damage and injuries. Moreover, more than 57,000 cases of non-fatal workplace violence were reported in the U.S. during the 2021–2022 period, significantly affecting the healthcare sector. Severe incidents of workplace violence have escalated, with numerous reports indicating substantial operational safety risks in 2022–2023. In densely populated regions like Egypt, approximately 51,900 fire accidents were recorded in 2020, exemplifying significant safety hazards *[6, 7, 8]*.

These challenges emphasize the pressing need for an intelligent surveillance system. Such a system should be capable of automatically processing video streams and sensor data, accurately detecting hazardous situations, and providing timely alerts. This advancement could significantly enhance safety and security across various environments, addressing the multifaceted threats faced in today's world *[9]*.

1.4. Project Purpose

The purpose of this project is to develop a smart, integrated surveillance and safety platform that can automatically detect and respond to security and environmental hazards in real time, reducing reliance on continuous human monitoring. The project aims to combine computer vision based video analytics with embedded sensor networks under a centralized backend architecture, where sensor events act as control signals that selectively activate the appropriate AI models only when risk indicators are present. By adopting this event-driven approach, the system targets improved detection speed, reduced false alarms through sensor video correlation, and efficient use of computational and energy resources. Ultimately, the project seeks to provide a scalable, secure, and context-aware solution capable of supporting critical use cases such as fire/smoke detection, gas leak detection, weapon and intrusion detection, and abnormal behavior recognition across environments including residential buildings, commercial facilities, industrial sites, healthcare institutions, and public infrastructures.

1.5. Project Scope

This project encompasses the systematic analysis, design, development, and validation of an intelligent surveillance and monitoring framework that integrates Artificial Intelligence, Computer Vision, and Embedded Systems. The scope is confined to the realization of a prototype-level system intended to demonstrate automated, context-aware surveillance through sensor-driven intelligence.

The project scope includes the following components:

1.5.1 Problem Analysis and Requirement Specification

The project involves analyzing traditional surveillance systems to identify limitations related to continuous monitoring and delayed threat detection, followed by defining the functional and non-functional requirements for an intelligent surveillance solution.

1.5.2 System Architecture and Design

The scope includes designing a centralized, modular architecture that integrates cameras, sensors, and AI modules using an event-driven control mechanism where sensor data triggers selective AI processing.

1.5.3 System Implementation

The project covers the implementation of embedded sensors and AI-based computer vision models for tasks such as intrusion detection, fire and smoke detection, face recognition, and abnormal behavior analysis, with sensor fusion applied to improve accuracy.

1.5.4 System Management and User Interface

An intelligent control dashboard is developed to enable real-time monitoring, system configuration, alert management, and secure user access.

1.5.5 Testing and Performance Evaluation

The system is tested under controlled scenarios to evaluate detection accuracy, response time, and computational efficiency, with comparisons to traditional surveillance approaches.

1.6. Objectives

1.6.1 Reduce Dependence on Continuous Human Monitoring

To provide an automated surveillance system that minimizes the need for constant human supervision by using AI-based detection and analysis.

1.6.2 Overcome Delayed and Missed Threat Detection

To enable real-time detection of critical events such as fire, gas leaks, intrusions, violence, and abnormal behavior, ensuring faster and more reliable responses.

1.6.3 Address Inefficient Continuous AI Processing

To solve the problem of unnecessary computational load by introducing a sensor-driven, event-based AI activation mechanism instead of continuous video analysis.

1.6.4 Improve Accuracy and Reduce False Alarms

To enhance detection reliability by fusing data from cameras and environmental sensors, reducing false positives common in traditional surveillance systems.

1.6.5 Handle Lack of Intelligent Behavioral Analysis

To provide advanced behavior and situation analysis, including crowd detection, physical altercation recognition, and restricted-area monitoring.

1.6.6 Solve Fragmented System Architecture Issues

To design a centralized backend that integrates video streams, sensor data, and AI engines into a unified and manageable platform.

1.6.7 Improve System Responsiveness and Resource Efficiency

To optimize system performance and energy consumption by executing AI models only when relevant events are detected.

1.6.8 Address Poor System Control and Management

To provide an intelligent dashboard for real-time monitoring, configuration of cameras and sensors, and automation rule management.

1.6.9 Ensure Secure and Controlled System Access

To solve security and privacy concerns by implementing encrypted communication and role-based access control for authorized users.

1.6.10 Support Scalability and Future Expansion

To overcome limitations of rigid surveillance systems by adopting a modular architecture that supports easy upgrades and deployment across different environments.

1.7. Success Criteria

1.7.1 End-to-End System Operation

A validated prototype performs the complete workflow from acquisition (camera/sensors) → event-driven model activation → AI inference → alert generation → logging → dashboard-based review.

1.7.2 Quantified Detection Performance

Each detection module is evaluated using standard metrics (e.g., precision, recall, F1-score, and false-positive rate), with results reported on representative test cases/datasets.

1.7.3 Real-Time Responsiveness

The system meets a documented response-time target by reporting measured end-to-end alert latency under typical operating conditions (e.g., sensor trigger to dashboard/mobile alert).

1.7.4 Efficiency Improvement Demonstration

Event-driven activation shows measurable reduction in compute usage relative to continuous inference (e.g., fewer inference calls, lower average CPU/GPU utilization, or reduced power consumption), supported by experimental logs.

1.7.5 Dashboard Functional Verification

The dashboard successfully supports configuration, monitoring, access control, alert handling, and event history retrieval, verified through defined functional test cases.

1.7.6 Security Controls Validation

Authentication and role-based access control are enforced correctly, and communication channels are protected using encryption, verified through configuration and test evidence.

1.7.7 Stability and Reliability

The prototype operates continuously for a defined test duration without critical failures, and all generated events remain retrievable and correctly indexed.

1.8. Report outline

This project report is structured as follows:

- **Chapter 1** provides an overview of the system, introducing the background, problem statement, project purpose, scope, objectives, success criteria, and the expected impact of the proposed smart surveillance system.
- **Chapter 2** presents a review of related work, discussing existing surveillance systems, identifying their limitations, and comparing them with the proposed solution.
- **Chapter 3** describes the methodology of the project, including system requirements, software engineering design, system architecture, implementation details, tools and technologies used, datasets, and AI models.
- **Chapter 4** presents the experimental results and system evaluation, including simulation outcomes, AI model performance results, and user interface design mockups.
- **Chapter 5** concludes the report by summarizing the project findings, discussing system limitations, and outlining future work and possible improvements. References and appendices follow this chapter.

CHAPTER 2

RELATED WORK

2. RELATED WORK

2.1. Existing Systems

The IBM Smart Surveillance System (S3) is one of the early systems designed for event-based video surveillance. It uses an open and flexible structure. The system has two main parts: the Smart Surveillance Engine (SSE) and the Middleware for Large Scale Surveillance (MILS). The SSE works as the front end and handles video analysis, while MILS manages data storage and queries.

The SSE uses plug-ins to connect different analysis tools. These tools perform tasks such as behavior analysis, license plate recognition, and face recognition in real time. The system creates event data in XML format and sends it to MILS using web services. MILS stores and indexes this data in a database, which allows users to search large amounts of surveillance data using SQL-like queries. The SSE can detect and track objects, classify them as people or vehicles, and generate alerts for events such as motion, abandoned objects, or camera tampering. The system can also combine events from multiple cameras to detect complex activities like shoplifting. This was tested in a retail store to detect return fraud by linking camera data with transaction records *[10]* .

In contrast, some studies focus on low-cost surveillance systems for small places such as homes and offices. One such system uses a Passive Infrared (PIR) sensor to detect motion and a Raspberry Pi to capture images using a camera. When motion is detected, the image is sent to the user by email through Wi-Fi. The system runs on a Raspberry Pi 3 using Python to control the sensor, camera, and other hardware. Tests show that the system can reliably capture images and send alerts, making it a low-cost and practical security solution *[11]* .

Recent research has also focused on improving public safety by using deep learning for weapon detection. One study presents an automated surveillance system that detects guns using the YOLO V3 model. The system uses transfer learning on a custom dataset and does not require powerful hardware. Results show that YOLO V3 performs better than older models like YOLO V2, with an accuracy of 98.89%. This high performance comes from its improved network design, which helps detect small objects. When a weapon is detected, the system records the time and location, alerts security staff, and saves the event in a database *[12]* .

To support devices with limited resources, other research proposes lightweight deep learning models. The MSD-CNN model is designed to detect abnormal objects such as guns and knives in surveillance videos. It also classifies these objects into specific categories. The model is trained using a large dataset created from existing image databases. Its two-branch design improves accuracy while keeping computation low. Experiments show that MSD-CNN performs better than models like YOLO and Faster R-CNN and can run in real time on a Raspberry Pi 4 *[13]*.

Face recognition is another important part of smart surveillance. A hybrid system called HC-CNN combines Haar Cascade and CNN techniques. Haar Cascade is used for fast face detection, while CNN is used for accurate face recognition. The system uses both edge devices and cloud computing. Simple tasks are handled locally to reduce delay, while complex tasks are processed in the cloud. Tests show that HC-CNN performs better than using CNN alone and works well in real time *[14]* .

Anomaly detection in crowded areas is also widely studied. One approach uses a deep learning model based on 3D convolution and ConvLSTM layers to learn normal behavior patterns in videos. The model is trained only on normal events. When unusual behavior occurs, it produces a high reconstruction error, which signals an anomaly. This method reduces the need for human monitoring but can be affected by changes in lighting and background [15] .

Other systems focus on behavior analysis using object tracking. One expert surveillance system for shopping malls uses background subtraction to detect people and corrects errors using blob fusion. It tracks movement using Kalman filtering and analyzes appearance features to handle occlusion. By studying movement paths, the system detects events such as loitering, shop entry, and unattended cash desks. Tests show good performance in real-time use [16] .

Smart surveillance is also used in smart homes. Systems like SmartMonitor combine video analysis with home automation. The system detects and tracks objects in real time and triggers alarms based on the situation. One important application is fall detection for elderly people. The system analyzes motion and body shape changes to detect falls. Experiments confirm that this method works well across different datasets [17] .

At a larger scale, smart cameras are becoming common in smart homes to improve security and automation. These systems support face recognition, object detection, and interaction with other smart devices. However, challenges still exist, such as privacy concerns, data security risks, false alarms, high costs, and poor performance in low-light conditions [18] .

Finally, large public surveillance systems focus on crowd monitoring and early incident detection. PublicVision is one such system that uses secure communication and centralized analysis. It uses encrypted networks to protect video data and a deep learning model based on the Swin Transformer to analyze crowd behavior. A new dataset is used to classify crowd size and violence levels. The system runs in real time using NVIDIA DeepStream and provides alerts with detailed information, showing the value of intelligent surveillance for public safety [19] .

2.2. Overall Problems of Existing Systems

Despite significant advancements in smart surveillance technologies, the reviewed systems still suffer from several critical limitations that reduce their effectiveness, scalability, and practicality in real-world environments. The main overall problems can be summarized as follows:

1. **Heavy Dependence on Continuous Video Processing**
   Most existing systems rely on continuous video analysis, where AI models run all the time regardless of actual risk presence.
   This leads to:
   - High computational cost
   - Increased energy consumption
   - Poor scalability when multiple cameras are deployed

   Even advanced systems (e.g., weapon detection, crowd analysis, anomaly detection) lack selective or event-driven activation mechanisms.

2. **Limited or No Sensor Integration**
   The majority of reviewed systems depend only on visual data:
   - PIR-based systems detect motion only
   - Vision-only AI models analyze frames without environmental context

   This causes:
   - High false alarm rates
   - Inability to confirm events using multi-source data
   - Weak detection in poor lighting, occlusion, smoke, or shadows

   Very few systems implement sensor fusion between cameras and environmental sensors.

3. **Narrow Functional Focus (Single-Threat Systems)**
   Many existing solutions are designed to detect only one type of threat, such as:
   - Weapon detection only
   - Face recognition only
   - Fall detection only
   - Crowd behaviour only

   As a result:
   - They cannot handle multi-hazard environments
   - They fail when simultaneous or compound threats occur (e.g., fire + intrusion)

4. **High False Positives and Context Misunderstanding**
   Several systems struggle with:
   - Lighting changes
   - Weather conditions
   - Shadows and reflections
   - Dynamic backgrounds
   - Occlusions

   Because they lack context-aware decision-making, they often:
   - Misclassify normal behaviour as abnormal
   - Trigger unnecessary alerts
   - Reduce trust in the system

5. **Limited Real-Time Responsiveness**
   Although many papers claim “real-time” performance:
   - Some systems suffer latency due to cloud-only processing
   - Others are constrained by low-power edge devices
   - Heavy models reduce response speed under load

   This limits their effectiveness in time-critical incidents such as violence, fire, or gas leaks.

6. **Poor Scalability and Deployment Complexity**
   Several systems:
   - Are tested only in controlled or small-scale environments
   - Do not scale efficiently to large camera networks
   - Require high-end hardware or stable internet connectivity

   This makes them unsuitable for:
   - Large facilities
   - Public infrastructure
   - Industrial or city-scale deployments

7. **Weak System-Level Integration**
   Many existing works focus on individual algorithms rather than a full system:
   - No centralized control
   - Limited user management
   - Weak integration between detection, alerting, and response

   This results in:
   - Fragmented solutions
   - Difficult system management
   - Poor operational usability

8. **Security and Privacy Concerns**
   Some systems:
   - Lack strong encryption
   - Depend heavily on cloud connectivity
   - Offer limited access control

   This raises concerns regarding:
   - Unauthorized access
   - Data leakage
   - Privacy violations

9. **Lack of Decision Support and Automation**
   Most systems:
   - Only detect events
   - Do not assist in decision-making
   - Provide alerts without prioritization or intelligent orchestration

   This increases the cognitive load on security personnel and delays response actions.

2.3. Comparison Between Existing and Proposed Method

Table 2.1: Comparison of methods

Feature / Aspect	Method A	Method B	Method C	Method D	Our Method
Monitoring Type	Manual human monitoring	Continuous AI video processing	Sensor-triggered basic capture	Continuous AI for specific task	Event-driven intelligent monitoring
Sensor Integration	None	Vision only	Limited (PIR only)	Mostly vision-based	Multi-sensor fusion (motion, gas, fire, etc.)
AI Processing Strategy	None	Always-on AI	No advanced AI	Task-specific AI only	Sensor-triggered selective AI activation
Threat Coverage	Very limited	Single threat type	Intrusion only	Single or narrow threat	Multi-hazard (fire, gas, intrusion, weapons, behaviour)
Context Awareness	None	Limited	None	Partial	High (sensor + vision correlation)
False Alarm Reduction	No	Limited	Poor	Moderate	High (sensor fusion & context)
Real-Time Response	Slow	Depends on load	Limited	Moderate	Fast, low-latency response
Scalability	Poor	Limited by compute	Poor	Moderate	Scalable centralized architecture
System Integration	Fragmented	Partial	Basic	Limited	Fully integrated system
User & Access Management	None	Basic	None	Limited	Role-based access control
Energy & Resource Efficiency	Very low	Low	Moderate	Low	High efficiency

As shown in Table 2.1, the proposed method overcomes the limitations of existing surveillance systems by integrating sensor-driven intelligence, multi-hazard detection, and context-aware AI processing within a scalable and secure centralized architecture.

CHAPTER 3

METHODOLOGY

3. METHODOLOGY

3.1. PROJECT DESCRIPTION

The Smart Surveillance and Safety System is a smart monitoring system that uses IoT sensors, cameras, and AI to detect unusual or dangerous events in real time. Unlike regular CCTV systems that record continuously and need people to watch them, this system can understand the environment and act automatically when something abnormal happens.

The system has four main parts physical, communication, intelligence, and application. The physical part includes cameras and IoT sensors. These sensors connect to small controllers like ESP32, Arduino, or Raspberry Pi. The communication part sends sensor data using MQTT, video streams using RTSP, and real-time alerts through WebSockets, all securely with encryption. The intelligence part contains AI models and decision-making logic that are only activated when sensors detect a problem. The Application part is a web and mobile dashboard where users can see live videos, alerts, playback, and manage settings.

3.1.1. IoT

In this project, IoT sensors play a central role in making the system smart and efficient. Rather than continuously analyzing video, IoT sensors continuously monitor the environment and detect any unusual conditions. When a sensor records an abnormal reading, such as a high gas level or sudden motion it sends an alert to the backend server. The system then activates only the relevant AI model to analyze the video feed. This approach has several advantages: it saves computation and energy by activating AI only when necessary, detects hazards early before they become visible on cameras, and reduces false alarms by combining sensor readings with video analysis.

Table 3.1 : Hardware Components

Sensor Type	Model	Purpose in the System	How it Works in the Project
Gas Sensor	MQ-2	Detects dangerous gases	Sends a signal when gas levels exceed a safe threshold; triggers AI to check video for leaks or fire.
Smoke Sensor	MQ-2	Detects smoke in the environment	Works with temperature sensors to activate fire detection AI models.
Temperature Sensor	DHT22	Detects abnormal heat or sudden temperature changes	Helps identify fire or overheating in machines; works together with smoke sensors.
Motion Sensor (PIR)	PIR Motion Sensor (HC-SR501)	Detects human or object movement	Activates object detection or behaviour recognition AI when movement is detected.
Sound Sensor	KY-038 / KY-037	Detects unusual or loud sounds	Helps detect fights, vandalism, or breaking events; triggers AI to verify the event.
Vibration Sensor	KY-038 / KY-037	Detects mechanical movements or tampering	Used to detect camera tampering or physical attacks on objects; triggers AI to confirm visually.

3.1.2. AI

Artificial Intelligence (AI) is employed to detect and recognize abnormal or suspicious activities in surveillance environments, focusing on events such as weapons detection such as knives and firearms, fire, smoke, or gas leaks detection, restricted-area violations using facial recognition with whitelist/blacklist, and a wide range of human activities. These activities are categorized into classes including abuse, arson, violent acts, theft, explosion, vandalism, camera tampering, normal behavior, and atypical movements. AI is not continuously active; rather, it is triggered by IoT sensors when unusual events are detected. Once activated, AI models analyze short video segments to detect objects, recognize faces, identify human actions or abnormal behaviors, and classify events. To develop these models, three main strategies are applied: (1) fine-tuning existing pre-trained models, such as YOLO for object detection, I3D for action recognition, and FaceNet/ArcFace for face recognition, on surveillance-specific datasets; (2) designing custom models from scratch for specialized tasks, including detecting complex abnormal behaviors or camera tampering; and (3) employing ensemble learning to combine multiple models, thereby improving accuracy and reducing false alarms.

For developing and evaluating the Intelligent Surveillance & Safety System, we used a combination of publicly available datasets and custom-labeled surveillance data to cover a wide range of abnormal and suspicious activities. The datasets include both video-based activity recognition and image-based detection tasks.

Table 3.2 : Dataset for Activity Recognition

Class	Total Samples	Train (70%)	Validation (15%)	Test (15%)	Source / Notes
Abuse	500	350	75	75	UCF-Crime + online videos
Arson	450	315	68	67	UCF-Crime + online videos
Violent Acts	600	420	90	90	UCF-Crime + online videos
Theft/Robbery	550	385	83	82	UCF-Crime + online videos
Explosion	400	280	60	60	UCF-Crime + online videos
Vandalism	450	315	68	67	UCF-Crime + online videos
Camera Tampering	400	280	60	60	UCF-Crime + online videos
Normal	600	420	90	90	UCF-Crime + online videos
Atypical Behaviors	550	385	83	82	UCF-Crime + online videos
Total	4,500	3,150	675	675	

Table 3.3: Dataset for weapon, fire, face recognition

Task / Application	Class	Total Samples	Train (70%)	Validation (15%)	Test (15%)	Source
Weapon Detection	Weapon	9,657	5,871	1,491	2,295	Hugging Face
	Non-Weapon					
Fire / Smoke Detection	Normal	3,600	2,520	540	540	Kaggle
	Fire					
	Smoke					
Restricted-Area / Face Recognition	Whitelist	2,800	1960	420	420	Hugging Face
	Blacklist					

3.2. System Requirements

3.2.1. Functional Requirements

1. The system shall provide a centralized backend server that manages data processing, AI inference, user authentication, and configuration.
2. The system shall expose API that allows applications (web, mobile, other devices) to connect to the system.
3. The system shall support WebSocket connections to deliver real-time video streams and alerts.
4. The system shall support MQTT protocols for receiving sensor data.
5. The system shall connect to cameras using the RTSP protocol for video ingestion.
6. The system shall provide real-time display of video streams from connected cameras.
7. The system shall overlay metadata (camera name, location, timestamp, status) on the live stream.
8. The system shall support digital zoom (in/out) on live video feeds.
9. The system shall provide PTZ (Pan/Tilt/Zoom) control for supported cameras.
10. The system shall support Night Vision mode when infrared is active.
11. The system shall decode video formats H.264 and H.265.
12. The system shall support Smoke, Gas, Motion, Temperature, Sound/Vibration sensors.
13. Smoke sensor events shall trigger the Fire/Smoke AI model and create a high-priority fire alert.
14. Gas leak events shall trigger hazard alerts and activate emergency exhaust fans.
15. Motion sensor events shall activate Object and Behavior Recognition AI models.
16. The system shall allow configuration of automation rules linking sensors to AI model activation .
17. The system shall detect weapons, fire, smoke, and human faces.
18. The system shall detect behaviors including fighting, intrusion, camera tampering, and restricted-area violations , abuse , arson, theft/robbery, explosion, vandalism.
19. The system shall classify detected faces as whitelist or blacklist using face recognition.
20. The system shall allow AI models to operate in continuous mode.
21. The system shall allow AI models to operate in sensor-triggered mode.

22. The system shall support AI model scheduling based on predefined time windows.

23. The system shall support continuous video recording for all cameras.
24. The system shall support event-based recording triggered by AI detections.
25. The system shall support motion-triggered recording using AI or sensors.
26. Authorized users shall be able to start manual (on-demand) recording.
27. The system shall allow administrators to define custom recording schedules.
28. The system shall generate real-time alerts for weapons, violence, fire/smoke, gas leak, intrusion, occlusion, restricted-area violation, blacklist detection, and door breach events.
29. Alerts shall be delivered via the Web Dashboard and Mobile App through push notifications.
30. Alerts shall be optionally delivered via email.
31. Alerts shall contain event type, timestamp, camera, location, and severity level.
32. Alerts shall include a snapshot of the event frame.
33. The system shall provide a “Forgot Password” functionality via email.
34. The dashboard shall support multi-camera live monitoring in a configurable grid layout.
35. The dashboard shall provide alert filtering by event type, camera, severity, and time.
36. Authorized users shall mark alerts as Resolved or Acknowledged.
37. The dashboard shall provide a searchable video archive with playback and download options.
38. The mobile app shall allow authenticated users to view live camera streams.
39. The mobile app shall receive push notifications for alerts.
40. Opening a notification shall show the alert with snapshot and metadata.

3.2.2. Non Functional Requirements

3.2.2.1. Performance

1. The system shall display live video streams with a latency of less than 2 seconds.
2. AI inference per video frame shall not exceed 3 seconds.
3. The system shall support at least 200 concurrent cameras and 500 sensors.
4. The system shall automatically reconnect a camera stream within 2 seconds after a connection loss.

3.2.2.2. Security

1. All communication shall be encrypted using HTTPS/TLS.
2. User passwords shall be stored using strong salted hashing (e.g., bcrypt).
3. The system shall be protected against SQL Injection and XSS attacks.
4. Role-Based Access Control (RBAC) shall be implemented (Admin, User).

3.2.2.3. Scalability

1. The system shall allow adding new cameras and sensors without redesigning the system.
2. The system shall be scalable to support up to 200 cameras and 500 sensors.

3.2.2.4. Reliability & Availability

1. The system shall have an annual availability of 99.9%.
2. The system shall handle network outages without data loss.
3. All events and alerts shall be logged and retrievable for at least 6 months.

3.2.2.5. Usability

1. The user interface (Web + Mobile) shall be intuitive and user-friendly.
2. Users shall be able to set up a new camera or sensor within 5 minutes.
3. Alerts shall be clear and understandable to the users.

3.2.2.6. Maintainability

1. The system shall allow maintenance and addition of new features with minimal downtime.
2. All code and APIs shall be fully documented.

3.2.2.7. Compatibility

1. The system shall support all IP cameras with RTSP and ONVIF protocols.
2. The web application shall work on modern browsers (Chrome, Edge, Firefox).

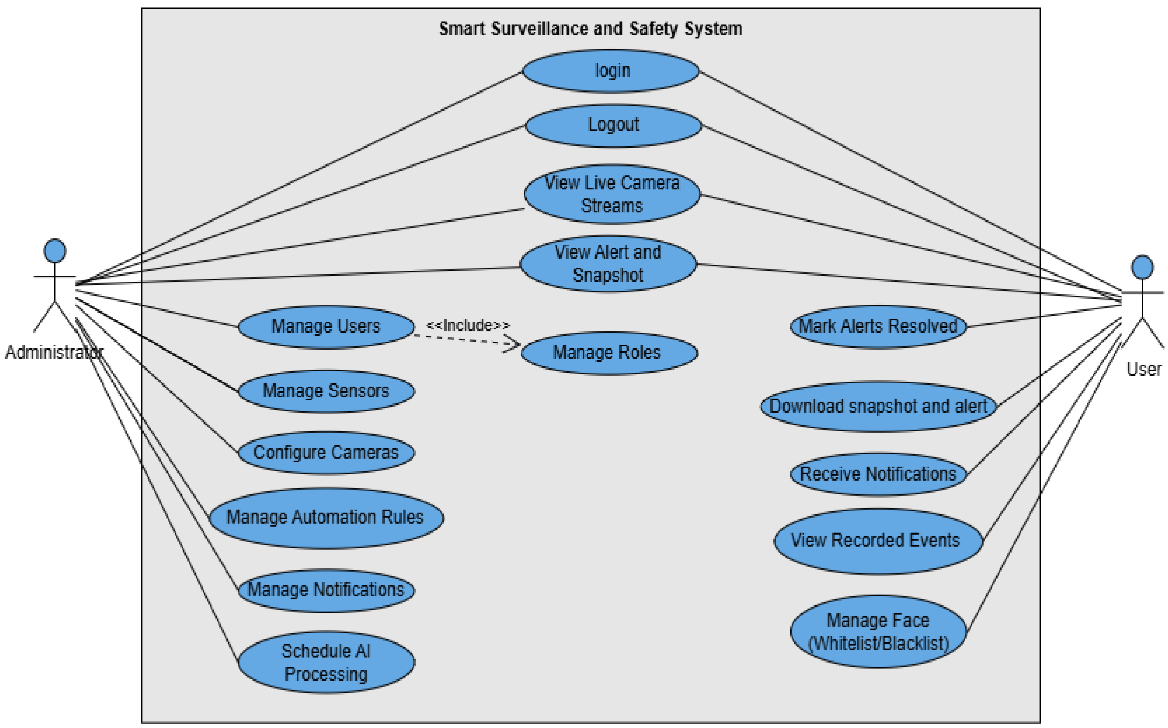
3.3. Software Engineering Design

3.3.1. System Use Case

3.3.1.1. Use Case Diagram

This diagram outlines all use cases in our system and their relations.

Figure 3.1: Use Case Diagram



3.3.1.2. Use Case Scenarios

1. Login

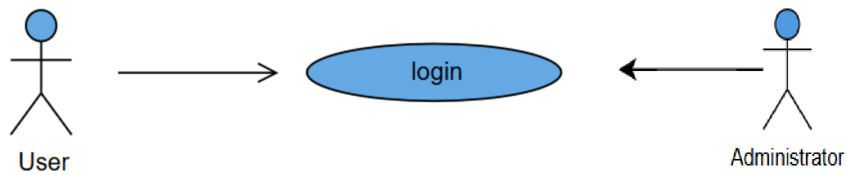


Table 3.4: Login Use Case

Use Case	**1**
Name	Login
Actor	User/Administrator
Description	The user/admin enters valid login credentials to gain access to the system
Priority	High
Precondition	1. The user/admin must have an existing registered account. 2. The system is online and reachable.
Postcondition	1. The user/admin is successfully logged in and granted access to system features. 2. The user's/admin's session is created.
Main flow	
User	**System**
1. The user enters username/email and password.	2. The system validates the credentials. 3. The system checks account status (active/not locked). 4. The system authenticates the user. 5. The system redirects the user to the main dashboard.
Alternative flow	
1. Invalid Credentials: a) The system detects wrong username or password. b) The system shows an error message. c) The user retries login. 2. Account Locked or Disabled: a) The system detects the account is locked. b) The system prevents login and displays a warning.	

2. Logout

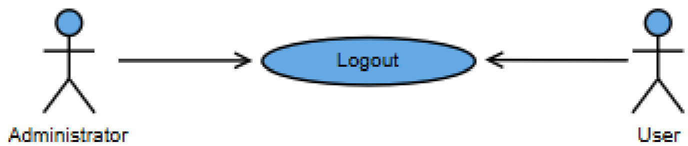
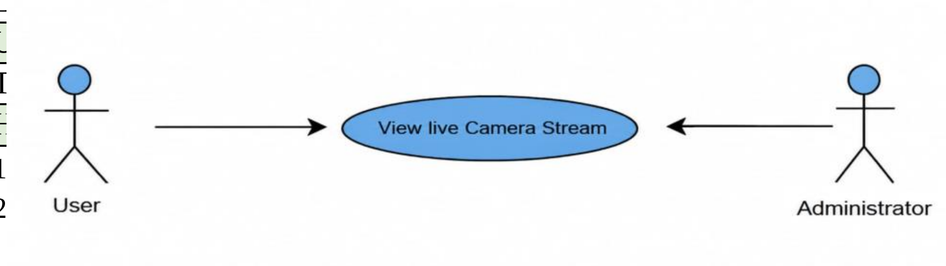


Table 3.5 : Logout Use Case

Use Case	**2**
Name	L
Actor	U
Description	T
Priority	H
Precondition	1 2
Postcondition	1. The user's/admin's session is successfully destroyed/invalidated. 2. The user is redirected to the login page or landing page.



Main flow	
User	**System**
1. The user clicks on the "Logout" button.	2. The system processes the logout request. 3. The system invalidates the current session and clears session data. 4. The system removes any stored authentication tokens (if applicable). 5. The system redirects the user to the login screen.

Alternative flow
1. **Session Already Expired:** a) The user clicks logout, but the session has already timed out. b) The system detects the expired session. c) The system redirects the user to the login page with a "Session Expired" message. 2. **Network Interruption:** a) The system fails to reach the server to invalidate the session. b) The system clears the local session/cache to ensure security on the client side. c) The system informs the user and redirects them to the login page.

3. View Live Camera Streams

Table 3.6 : View Live Camera Streams Use Case

Use Case	**3**
Name	View Live Camera Streams
Actor	User/Administrator
Description	The user/admin views real-time video streams from available cameras.
Priority	High
Precondition	1. The user must be authenticated. 2. At least one camera is online.
Postcondition	Live stream is displayed to the user.

Main flow	
User	**System**
1. The user navigates to the live stream section. 3. The user selects a camera.	2. The system loads available cameras. 4. The system retrieves and displays the live video feed.

Alternative flow
1. Camera List Loading Failure a) The system fails to load the list of available cameras. b) The system shows a message: "Unable to load cameras." c) User may retry. 2. Camera Offline a) The system detects the selected camera is offline. b) The system displays "Camera Offline".

4. View Alert and Snapshot

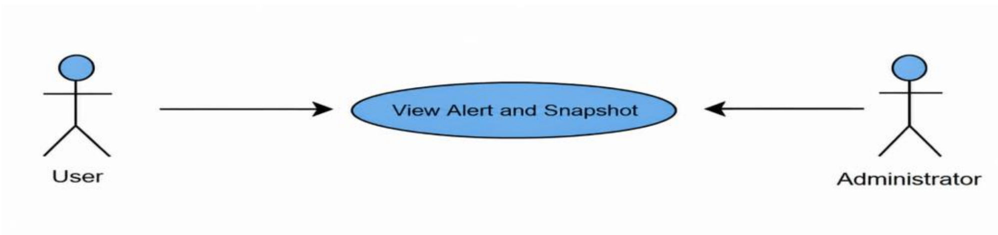


Table 3.7: View Alert and Snapshot Use Case

Use Case	5	
Name	View Alert and Snapshot	
Actor	User/Administrator	
Description	User/Admin reviews generated alerts along with the snapshot captured during the event.	
Priority	High	
Precondition	1. The user must be authenticated. 2. Alerts must already exist in the system.	
Postcondition	Alert details and snapshot are displayed.	
Main flow		
User		System
1. The user opens the Alert Center. 3. The user selects a specific alert		2. The system loads recent alerts. 4. The system displays alert information and the associated snapshot.
Alternative flow		
1. Snapshot Not Available a) If the system can't retrieve the snapshot, the system displays alert details only with a warning "Snapshot missing".		

5. Mark Alert Resolved

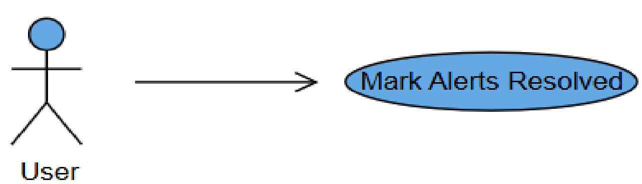


Table 3.8 : Mark Alert Resolved Use Case

Use Case	6	
Name	Mark Alerts Resolved	
Actor	User	
Description	The user marks an alert as resolved after reviewing it.	
Priority	Medium	
Precondition	1. The user must be authenticated. 2. The Alert must exist and be in "Pending" state.	
Postcondition	1. The state of alert is changed to "Resolved". 2. System logs resolution time.	
Main flow		
User		**System**
1. The user selects an alert. 2. The user clicks "Mark as Resolved".		3. The system updates the alert status. 4. The confirmation message appears.
Alternative flow		
1. Alert Already Resolved a) If the system detects the alert is already resolved, the system prevents duplicate action and informs the user.		

6. Download Snapshot and Alert

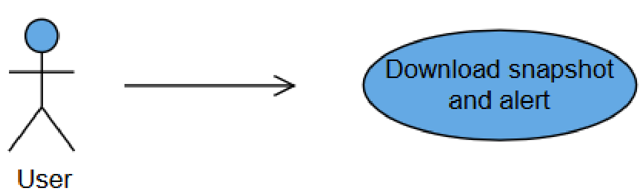


Table 3.9 : Download Snapshot and Alert Use Case

<table>
<tr><th>Use Case</th><th colspan="2">7</th></tr>
<tr><td>Name</td><td colspan="2">Download Snapshot and Alert</td></tr>
<tr><td>Actor</td><td colspan="2">User</td></tr>
<tr><td>Description</td><td colspan="2">The user downloads a snapshot image and its corresponding alert details in a downloadable format.</td></tr>
<tr><td>Priority</td><td colspan="2">Medium</td></tr>
<tr><td>Precondition</td><td colspan="2">1. The user must be authenticated.
2. The alert must contain downloadable data.</td></tr>
<tr><td>Postcondition</td><td colspan="2">File is downloaded to the user’s device.</td></tr>
<tr><th colspan="3">Main flow</th></tr>
<tr><th colspan="2">User</th><th>System</th></tr>
<tr><td colspan="2">1. The user opens an alert.
2. The user clicks “Download”.</td><td>3. The system prepares the alert data and snapshot.
4. The system sends file to the user for download.</td></tr>
<tr><th colspan="3">Alternative flow</th></tr>
<tr><td colspan="3">1. File Generation Error
a) If the system encounters an error while generating the file, the system notifies the user and suggests retrying.</td></tr>
</table>

7. Receive Notifications

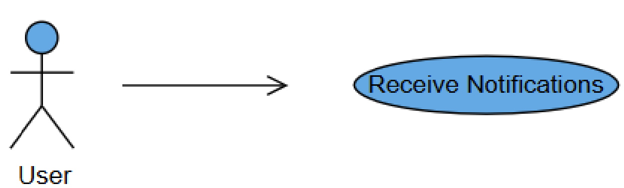


Table 3.10 : Receive Notification Use Case

Use Case	**8**
Name	Receive Notifications
Actor	User
Description	The system sends real-time notifications to the user about alerts or system events.
Priority	High
Precondition	1. The user must be authenticated and have push notifications enabled. 2. An alert/event must be generated.
Postcondition	The user receives a system notification.
Main flow	
User	**System**
4. The user views the notification.	1. The Camera or the AI module detects an event. 2. The system generates an alert. 3. The notification is pushed to the user (mobile app, email, dashboard).
Alternative flow	
1. Notification Delivery Failed a) If a network error prevents delivery, the system retries sending or logs the failure.	

8. View Recorded Events

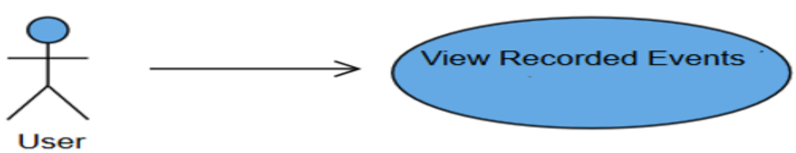


Table 3.11 : View Recorded Events / Playback Use Case

Use Case	**9**
Name	View Recorded Events / Playback
Actor	User
Description	The user views stored video recordings and plays back previous events
Priority	Medium
Precondition	1. The user must be authenticated. 2. Recordings must exist in storage.
Postcondition	Recorded event or playback video is displayed.
Main flow	
User	**System**
1. The user navigates to the "Recordings" page. 3. The user selects a specific recording.	2. The system loads available events and recordings. 4. The system loads the video and starts playback.
Alternative flow	
1. Recording Not Found a) If the system can't locate the video file, the system displays an error message.	

9. **Manage Recording**

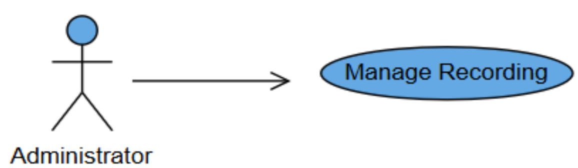


Table 3.12 : Manage Recording Use Case

Use Case	**10**
Name	Manage Recording

Actor	Administrator
Description	Allows the Administrator to define recording parameters such as quality, retention period, and triggers (continuous, event-based) for specific cameras or zones.
Priority	High
Precondition	1. The Administrator is authenticated. 2. The selected camera(s) and storage solution are active and configured.
Postcondition	New recording rules are successfully applied, and the system confirms the enforcement of the new data retention policy.

Main flow	
Administrator	**System**
1. The admin navigates to the "Recording Settings" section in the dashboard. 2. The admin selects a specific camera or a zone to configure. 4. The admin modifies the recording quality and retention period . 5. The admin sets the recording trigger (Continuous or Event-based). 6. The admin clicks the "Apply/Save Settings" button.	3- The system displays the current recording parameters (Quality, Retention, Triggers). 7- The system validates the new settings against available storage capacity. 8- The system updates the recording rules and displays a success confirmation message.

Alternative flow
1. Storage Limit Exceeded: a) The system detects that the requested quality/retention period exceeds the available storage capacity. b) The system displays a warning message. c) The admin adjusts the parameters to fit the storage limit. **2. Camera Connection Lost:** a) The admin attempts to save settings for a camera that has gone offline. b) The system displays an error message: "Camera unreachable." c) The system saves the configuration to be applied once the camera is back online.

10. Manage Face (Whitelist / Blacklist)

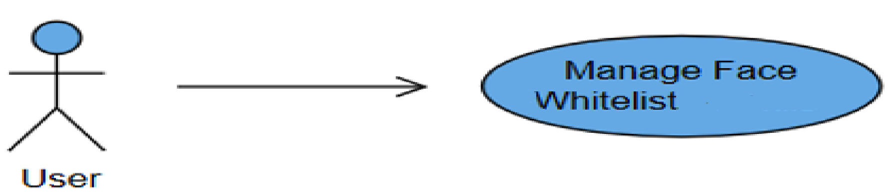


Table 3.13 : Manage Face (Whitelist / Blacklist) Use Case

<table>
<tr><th>Use Case</th><th colspan="2">11</th></tr>
<tr><td>Name</td><td colspan="2">Manage Face (Whitelist/Blacklist)</td></tr>
<tr><td>Actor</td><td colspan="2">User</td></tr>
<tr><td>Description</td><td colspan="2">Manages the database of authorized and unauthorized face profiles used by the AI recognition module to generate security alerts or suppress notifications.</td></tr>
<tr><td>Priority</td><td colspan="2">High</td></tr>
<tr><td>Precondition</td><td colspan="2">The Facial Recognition module is active and initialized.</td></tr>
<tr><td>Postcondition</td><td colspan="2">A new face profile (or modification to an existing one) is successfully stored and actively used by the AI engine.</td></tr>
<tr><th colspan="3">Main flow</th></tr>
<tr><th colspan="2">Administrator</th><th>System</th></tr>
<tr><td colspan="2">1. The User accesses Face Recognition Management.
2. Selects "Add New Profile."
3. Uploads an image and inputs metadata (Name, Role).
4. Assigns the profile as "Whitelist" or "Blacklist."
5. Clicks "Save Profile."</td><td>6. The system extracts facial vectors and stores them in the active AI database.
7. The system confirms the profile status and classification.</td></tr>
<tr><th colspan="3">Alternative flow</th></tr>
<tr><td colspan="3">1. Poor Image Quality:
a) If the uploaded image does not meet the clarity threshold for feature extraction, the system rejects the image and prompts the Administrator to upload a higher-quality one.</td></tr>
</table>

11. Manage Automation Rules

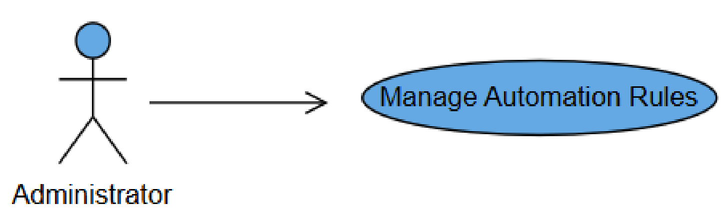


Table 3.14 : Manage Automation Rules Use Case

Use Case	**12**
Name	Manage Automation Rules
Actor	Administrator
Description	Allows the Administrator to create, modify, and delete automated IF-THEN rules that link a system event (IF) to a specific action (THEN), such as triggering an alarm or sending a notification.
Priority	High
Precondition	1. The Administrator is authenticated. 2. All required sensors, cameras, and notification methods are configured.
Postcondition	A new automation rule is successfully validated, activated, and stored in the system's runtime engine.
Main flow	
Administrator	**System**
1. The Administrator selects "Manage Automation Rules." 2. Selects "Create New Rule." 3. Defines the trigger source such as Camera3 and the condition such as "If Unauthorized Face Detected". 4. Selects one or more actions such as "Send Critical Push Notification" or "Activate Local Siren". 5. Clicks "Activate Rule."	6. The system validates the rule syntax and checks for resource conflicts. 7. The system activates the rule and displays a confirmation.
Alternative flow	
1. Rule Conflict: a) If the new rule conflicts with an existing high-priority rule, the system displays a warning detailing the conflict and prevents activation until the conflict is resolved.	

12.Manage Notifications

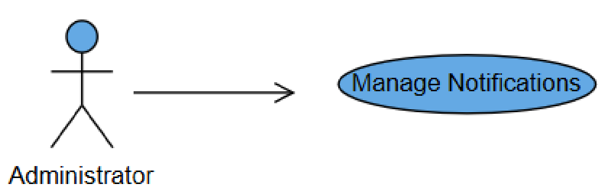


Table 3.15 : Manage Notifications Use Case

Use Case	**13**
Name	Manage Notifications
Actor	Administrator
Description	Customizes the delivery method (Push, Email, SMS), recipient roles, and priority level for various system events.
Priority	Medium
Precondition	User roles and contact information are accurately registered in the system.
Postcondition	The system's notification configuration is updated based on the new preferences.

Main flow	
Administrator	**System**
1. The Administrator accesses Notification Management. 2. Selects the event type to customize such as "Camera Disconnection". 3. Specifies the recipient roles and configures the delivery channels such as Push for Admins, or Email for all Users. 4. Sets the priority level (Normal/Critical). 5. Clicks "Save Preferences."	6. The system updates the alert routing tables. 7. The system confirms the saved changes.

Alternative flow
1. Channel Failure: a) If an external channel configuration (like SMS gateway) fails during setup, the system alerts the Administrator and defaults to Push/Email until the external channel is fixed.

12. Schedule AI Processing

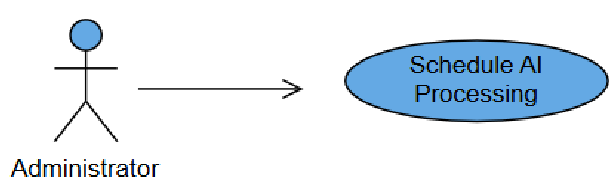


Table 3.16 : Schedule AI Processing Use Case

Use Case	**14**	
Name	Schedule AI Processing	
Actor	Administrator	
Description	Schedules complex analytical tasks -like object tracking or anomaly detection- to run once or recurrently on archived video or live feeds.	
Priority	High	
Precondition	The necessary AI processing module is available and licensed.	
Postcondition	The AI task is successfully queued for execution at the specified time, reserving the required computational resources.	
Main flow		
Administrator		**System**
1. The Administrator accesses the AI Scheduling dashboard.		
2. Selects the AI task type like "Object Search".		
3. Defines the video source (Archived or Live Feed, specifying camera/time range).		
4. Sets the execution schedule (Once/Recurring) and time.		
		5. The system verifies resource availability for the requested time.
6. The Administrator clicks "Confirm Schedule."		
		7. The system places the job in the processing queue.
Alternative flow		
1. Resource Overload: a) If the resource check (Step 5) fails due to peak load, the system suggests the next available time slot r asks the Administrator to run the task with lower priority.		

13. Manage Users

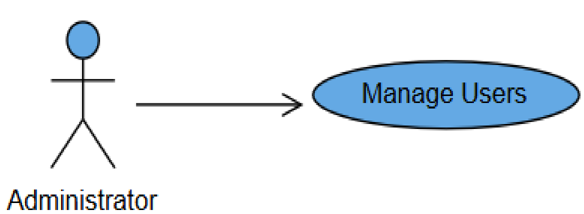


Table 3.17 : Manage Users Use Case

Use Case	**15**	
Name	Manage users.	
Actor	Administrator	
Description	Allows the Administrator to create, update, and delete user accounts within the system.	
Priority	High	
Precondition	The admin is authenticated and authorized to manage accounts.	
Postcondition	User account is successfully created, modified, or removed, and the user database is updated.	
Main flow		
Administrator		**System**
1. The admin navigates to the "User Management" dashboard. 3. The admin selects "Add New User". 5. The admin enters the new user's details and clicks "Submit". 8. The admin can also select an existing user to edit their details or delete their account.		2. The system displays a list of all existing user accounts. 4. The system provides a form for (Email, Username, Role, and Password). 6. The system validates the input and checks if the email/username is unique. 7. The system creates the account and sends a verification link to the **new user's email**. 9. The system updates the records and shows a success confirmation message.
Alternative flow		
1. Email/Username Already Exists: a) The system detects that the email or username is already registered. b) The system displays an error: "User already exists." c) The admin enters a different email or cancels the operation. **2. Verification Link Expired (New User side):** a) The new user attempts to verify their account but the link has expired. b) The system allows the admin to resend the verification email from the dashboard.		

14. Manage Roles

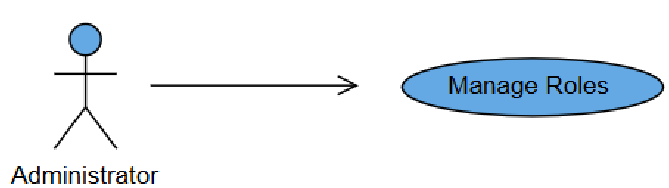


Table 3.18 : Manage Roles Use Case

Use Case	**16**
Name	Manage roles.
Actor	Administrator
Description	Distribution of roles and responsibilities.
Priority	High
Precondition	1. Admin login as admin. 2. Admin open dashboard and start creating users accounts.
Postcondition	The admin distributes the roles and responsibilities.

Main flow	
Administrator	**System**
1. The admin login to the app. 2. The admin start add/delete/update users' accounts. 3. The admin open dashboard and Distributes access privileges.	4. The user logins to their accounts and start check their privileges. 5. The app navigates to the user profile page.

Alternative flow
1. Invalid login a) If wrong email, the app sends wrong email message. b) If wrong password, the app sends wrong password message. 2. Existing user email: a) may the username unavailable to new create.

15. Manage Sensors

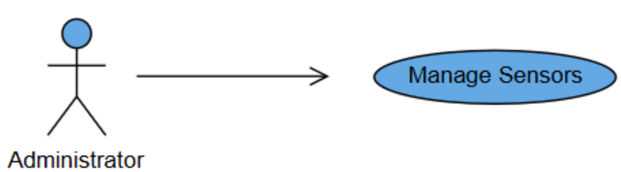


Table 3.19 : Manage Sensors Use Case

<table>
<tr><th>Use Case</th><th colspan="2">17</th></tr>
<tr><td>Name</td><td colspan="2">Manage sensors.</td></tr>
<tr><td>Actor</td><td colspan="2">Administrator</td></tr>
<tr><td>Description</td><td colspan="2">Connect the sensors with the app.</td></tr>
<tr><td>Priority</td><td colspan="2">High</td></tr>
<tr><td>Precondition</td><td colspan="2">1. the admin shall connect the sensors to the app by Wi-Fi.
2. The admin shall connect the AI model with sensors.</td></tr>
<tr><td>Postcondition</td><td colspan="2">Sensors are successfully linked to the system and ready to send data to the AI model.</td></tr>
<tr><td colspan="3">Main flow</td></tr>
<tr><td colspan="2">Administrator</td><td>System</td></tr>
<tr><td colspan="2">1. The admin navigates to the "Sensor Configuration" page.

3. The admin selects the detected sensor from the list.

5. The admin assigns a specific AI model (e.g., Object Detection) to the sensor.

7. The admin sets the detection sensitivity and saves settings.</td><td>2. The system scans for available sensors in the network.

4. The system establishes a secure Wi-Fi connection with the sensor.

6. The system validates the compatibility between the sensor and the AI model.

8. The system confirms the connection and displays a "Sensor Active" status.</td></tr>
<tr><td colspan="3">Alternative flow</td></tr>
<tr><td colspan="3">1. Sensor Not Found (Connection Issue):
a) The system fails to detect any sensors during the scan.
b) The system displays an error: "No sensors detected. Please check your Wi-Fi connection."
c) The admin retries the scan or checks the hardware.
2. AI Model Incompatibility:
a) The admin attempts to link an AI model that is not supported by the sensor type.
b) The system displays a warning message.
c) The admin selects a compatible AI model.</td></tr>
</table>

16. Configure Cameras

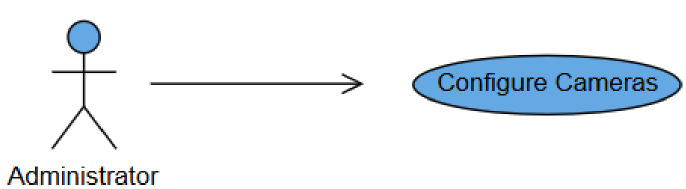


Table 3.20 : Configure Cameras Use Case

<table>
<tr><th>Use Case</th><th colspan="2">18</th></tr>
<tr><td>Name</td><td colspan="2">Configure Cameras.</td></tr>
<tr><td>Actor</td><td colspan="2">Administrator</td></tr>
<tr><td>Description</td><td colspan="2">The cameras shall share live stream video and recorded video in app by Wi-Fi.</td></tr>
<tr><td>Priority</td><td colspan="2">High</td></tr>
<tr><td>Precondition</td><td colspan="2">Cameras must be connected with app by Wi-Fi.</td></tr>
<tr><td>Postcondition</td><td colspan="2">The app shall show and record the videos that recorded by the cameras.</td></tr>
<tr><th colspan="3">Main flow</th></tr>
<tr><th colspan="2">Administrator</th><th>System</th></tr>
<tr><td colspan="2">1. The admin must login to dashboard.
2. Then connect cameras with app by Wi-Fi.</td><td>3. The camera start recording live video and sharing it with app.
4. The app shows the live records to the users that can access it.
5. The system shall store the records.</td></tr>
<tr><th colspan="3">Alternative flow</th></tr>
<tr><td colspan="3">1. Connection Timeout (Wi-Fi Issue):
a) The system tries to connect to the camera but the response takes too long.
b) The system displays an error: "Connection Timeout. Please check camera power or Wi-Fi signal."
c) The admin retries the connection.
2. Invalid Camera Credentials:
a) The admin enters the wrong password for the camera hardware itself.
b) The system displays: "Authentication failed for this device."
c) The admin re-enters the correct camera credentials.
3. Storage Full:
a) The system attempts to start recording but finds no space in the storage.
b) The system notifies the admin and stops recording until space is cleared.</td></tr>
</table>

3.3.2. Activity Diagram

This activity diagram shows the AI detection process in the surveillance system. Live camera streams are decoded and analyzed using AI models to detect objects or suspicious behavior. When a threat is identified, the system generates an alert, captures a snapshot, stores the alert in the database, and sends notifications to the dashboard and mobile devices, while continuously monitoring the video stream.

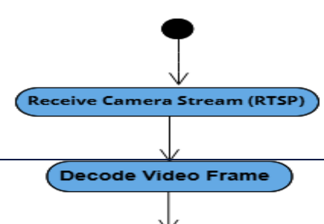


Figure 3.2 : AI Detection & Alert Generation Flow

This activity diagram illustrates the complete workflow of the monitoring system from user login to logout. It starts with user authentication, where credentials are validated before loading user preferences, the dashboard interface, and live camera streams. The system allows the user to view multiple cameras, open a single camera for detailed monitoring, and perform PTZ (pan, tilt, zoom) controls when supported. In addition, the user can manage alerts by viewing details and marking them as resolved, as well as access the recordings archive to stream or download video files. The process ends when the user logs out, at which point the session is destroyed and active connections are closed, ensuring proper system cleanup and security.

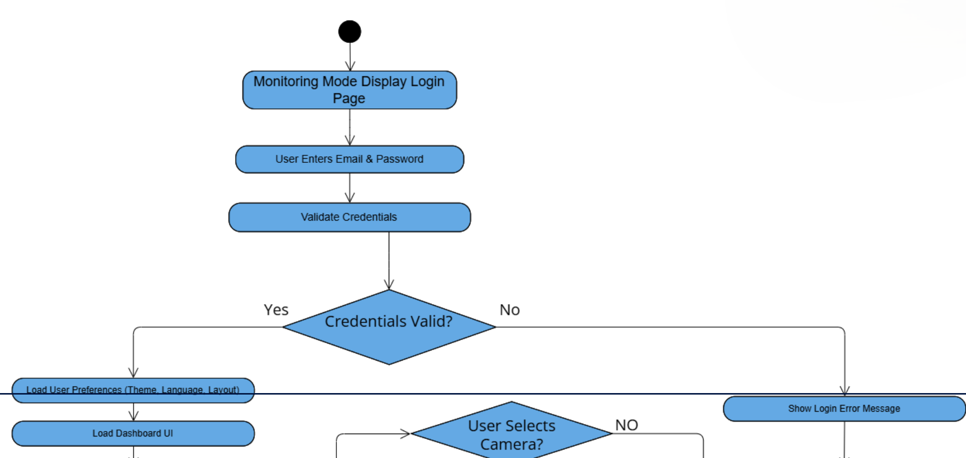


Figure 3.3 : User Monitoring & Interaction Flow

This activity diagram shows the sensor-based monitoring flow. Sensor data is evaluated against rules, optionally analyzed by AI, and used to detect hazards. When a risk is identified, automation actions are triggered, alerts are created and stored, and notifications are sent; otherwise, the system continues normal monitoring.

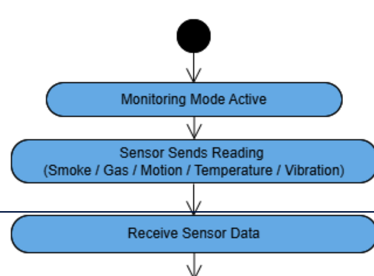


Figure 3.4 : Sensor Trigger & Automation Flow

3.3.3. Class Diagram

This UML Class Diagram illustrates the architecture of a Smart Surveillance or Security System. It defines four core entities: Users (administrators or operators who manage the system and resolve issues), Cameras (devices capable of streaming, recording, and PTZ movement), Sensors (hardware that detects specific environmental events), and Alerts (critical incidents generated by the devices). The relationships emphasize a workflow where Cameras and Sensors trigger Alerts (such as motion detection or hardware faults), which are then notified to and resolved by Users via dashboards or external messaging like Email and SMS.

Figure 3.5 : Class Diagram

3.3.4. ER Diagram

This Entity Relationship Diagram (ERD) outlines the database schema for the surveillance system, focusing on the relationship between users, devices, and security incidents. The model consists of five key entities: Admin, User, Camera, Sensor, and Alert. The relationships define a clear workflow where Cameras and Sensors detect activity and "Trigger" Alerts (logging details such as timestamp, severity, and status). These alerts are then linked to Users via a "Resolves" relationship, ensuring accountability for every incident. Additionally, an administrative hierarchy is established where Admins "Manage" multiple Users, controlling access and oversight within the system.

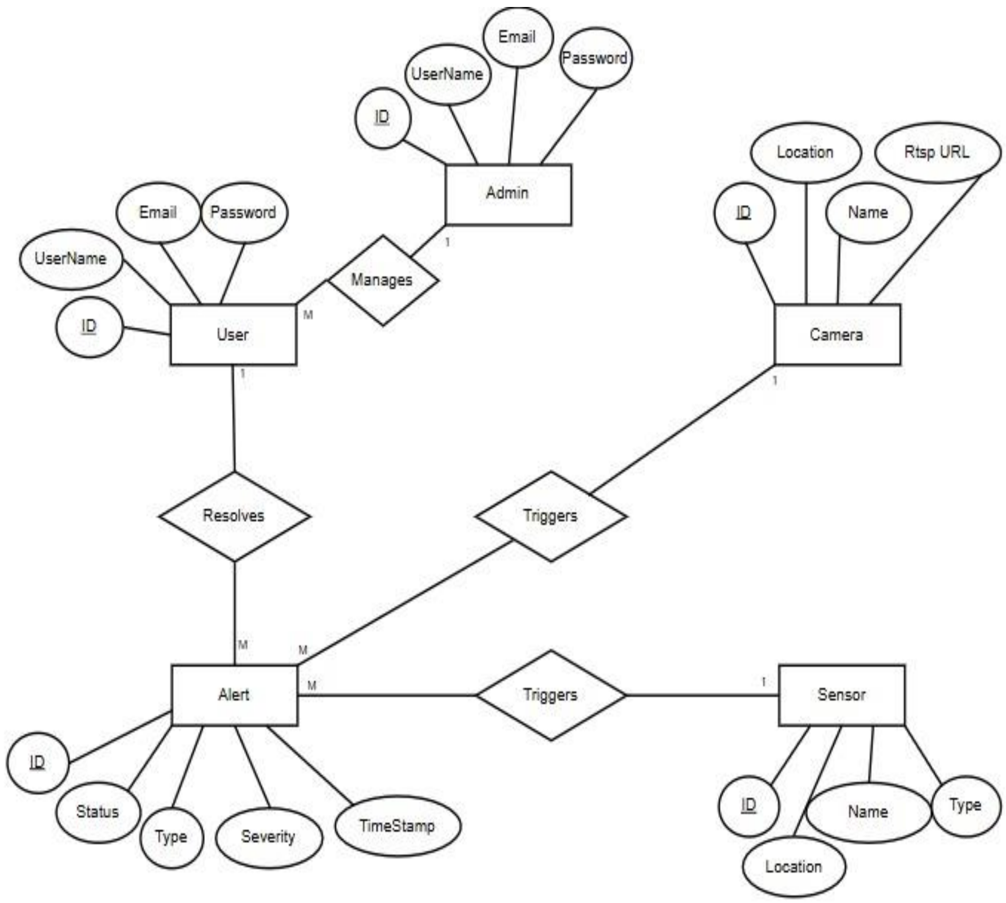


Figure 3.6 : ER Diagram

3.3.5. Business Model

This Business Model Canvas outlines the strategic framework for the AI-Based Smart Surveillance Mobile Application. The core Value Proposition centers on delivering real-time AI threat detection, instant mobile alerts, and automated incident responses to reduce reliance on manual monitoring. The business targets a broad range of Customer Segments, including industrial factories, shopping malls, and educational institutions, generating revenue through Subscription-based plans and enterprise licensing. To sustain operations, the model relies on Key Partners such as IP camera manufacturers and cloud service providers, while the Cost Structure is primarily driven by cloud infrastructure, GPU-enabled AI processing, and secure data storage.

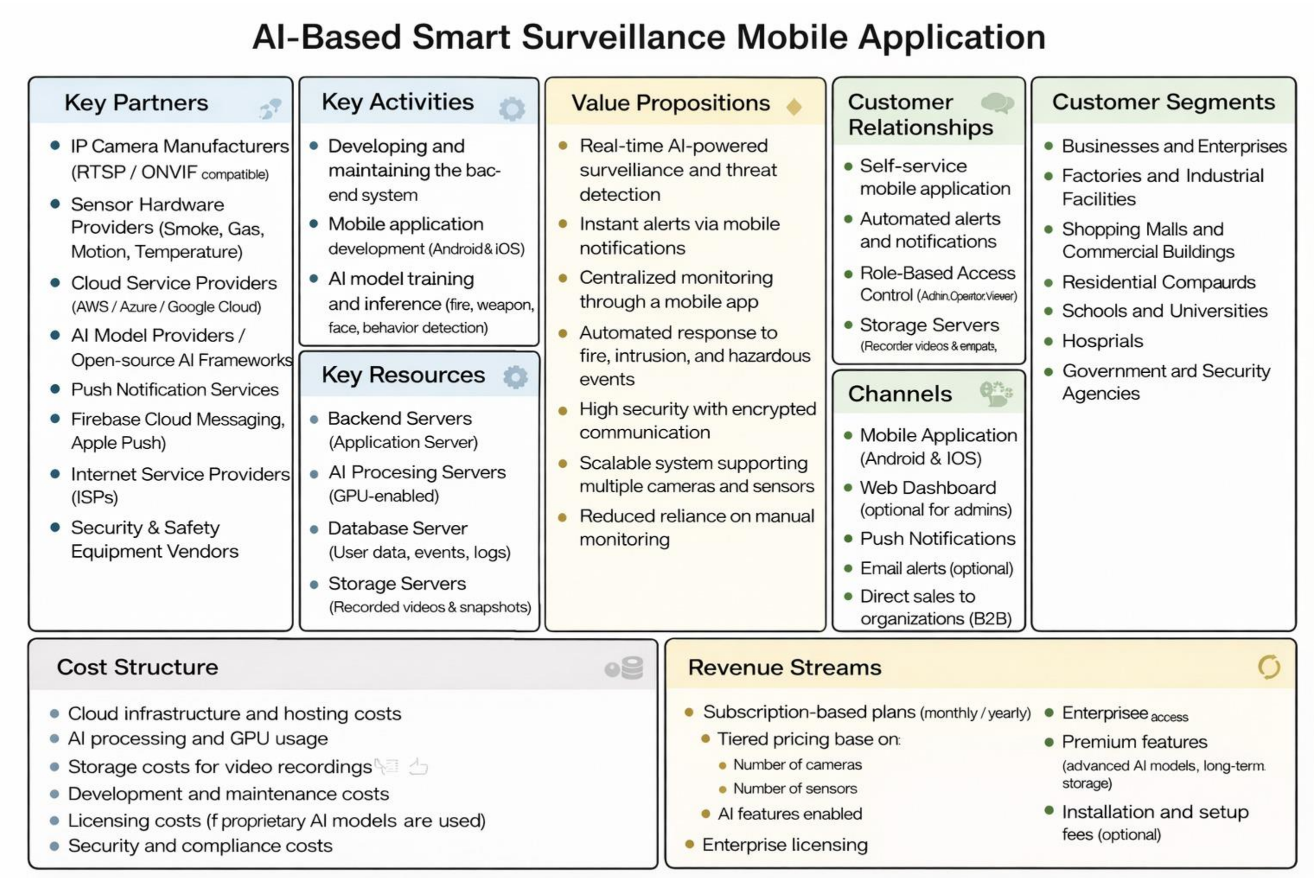


Figure 3.7 : Business Model

3.3.6. Block Diagram

This Block Diagram illustrates the high-level functional architecture of the Smart Surveillance System, organized into four primary blocks. The workflow begins at the Field Layer, where edge devices like IoT Sensors and Video Capture Units collect raw data. This input is transmitted to the Core System block, where a Secure Gateway and Media Streaming Server distribute the feed for live

viewing and simultaneous processing by the AI Analysis Engine. The results are stored in the Data Layer (handling Video Archives and Metadata), while the User Interfaces block ensures that real-time alerts and analytics are delivered to the Mobile App and Admin Dashboard.

Figure 3.8 : Block Diagram

3.3.7. Sequence Diagram

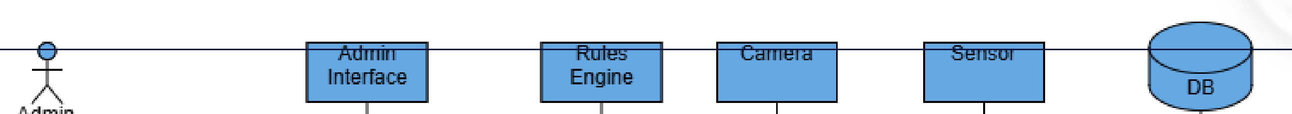
1. User

This Sequence Diagram details the end-to-end interaction flow for a user within the surveillance system. The process begins with Authentication, where the user validates credentials against the database to access the dashboard. For Live Monitoring, the app retrieves camera metadata and establishes a real-time video stream via a WebSocket connection bridged to the camera's RTSP feed. Concurrently, the AI Engine analyzes video frames; upon detecting a threat, it triggers an automated workflow that creates an alert, saves it to the database, and sends an immediate Push Notification to the user. Finally, the diagram illustrates the Incident Response phase, where the user retrieves alert details and updates the status to "Resolved" after addressing the issue.

Figure 3.9 : User Sequence Diagram

2. Admin

This Sequence Diagram illustrates the administrative workflows for configuring and managing the surveillance system. The process begins with Authentication, where the admin's credentials and role are validated against the database to grant access to the dashboard. Once logged in, the admin performs critical setup tasks, including Configuring Cameras (applying settings directly to the hardware) and Adding Sensors to the network. The diagram also highlights the interaction with the



Rules Engine to validate and activate automation protocols, as well as the User Management function, where changes to user accounts and system rules are committed to the database.

3.3.8. Deployment Diagram

Figure 3.10 : Admin Sequence Diagram

This deployment diagram shows a smart monitoring system where a mobile Android app communicates securely with backend services using HTTPS and WebSocket. The app uploads files (such as videos or images) to an AI Processing Server, which runs fire/smoke and weapon-detection models. Results and media are stored in a database and video/snapshot storage, while a recording service and alert service handle continuous recording and real-time notifications. The system also integrates AI integrity and MQTT sensors for live data input, enabling fast detection, storage, and alerting in one connected architecture.

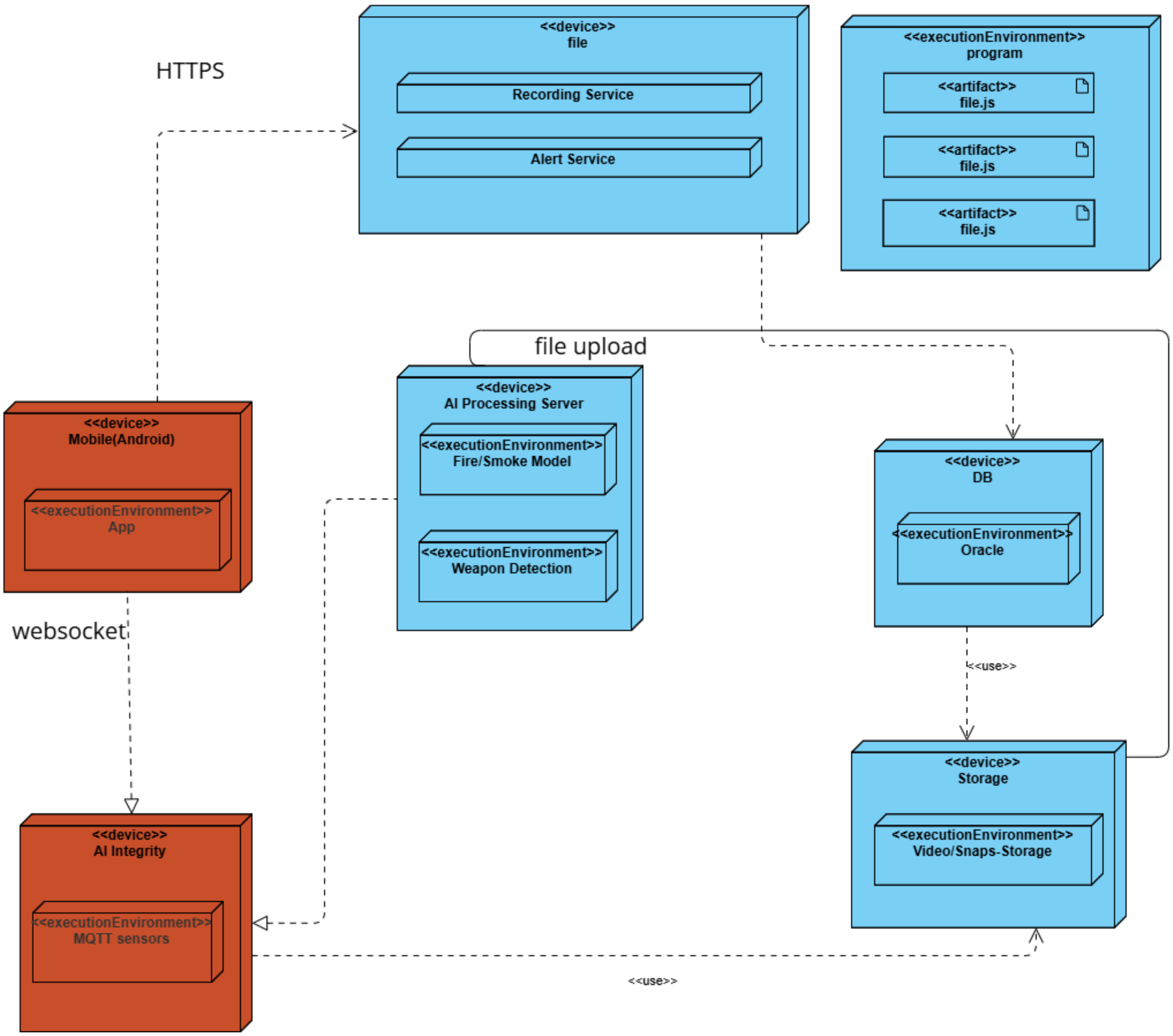
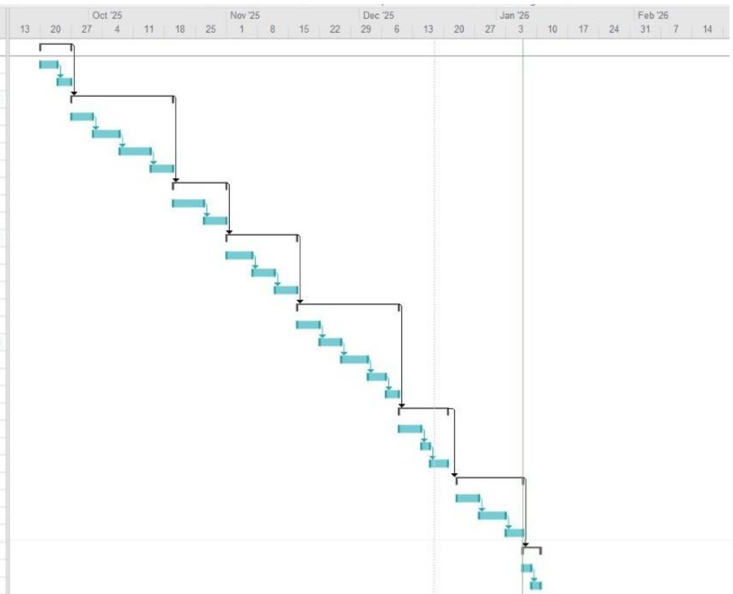


Figure 3.11 : Deployment Diagram

3.4. Project Plan

This project plan defines the structured roadmap for developing the Smart Surveillance System, outlining the project phases, tasks, durations, and dependencies from initial concept to final implementation and testing. The plan is organized to ensure a logical and continuous workflow, with clearly defined milestones that support effective scheduling, resource coordination, and progress tracking. By following this plan, the project aims to minimize risks, maintain timeline control, and ensure the successful delivery of a functional and reliable system within the specified academic timeframe.

	WBS	Task Name	Duration	Start	Finish
1	1	**Phase 1: Idea & Concept**	**7 days**	**Sat 9/20/25**	**Fri 9/26/25**
2	1.1	Finding the Project Idea	4 days	Sat 9/20/25	Tue 9/23/25
3	1.2	Writing the Abstract	3 days	Wed 9/24/25	Fri 9/26/25
4	2	**Phase 2 : Problem & Requirements**	**23 days**	**Sat 9/27/25**	**Sun 10/19/25**
5	2.1	Problem Formulation	5 days	Sat 9/27/25	Wed 10/1/25
6	2.2	Requirement Analysis	6 days	Thu 10/2/25	Tue 10/7/25
7	2.3	Writing Functional Requirement	7 days	Wed 10/8/25	Tue 10/14/25
8	2.4	Writing Non-Functional Requirement	5 days	Wed 10/15/25	Sun 10/19/25
9	3	**Phase 3 : Literature & Related Work**	**12 days**	**Mon 10/20/25**	**Fri 10/31/25**
10	3.1	Searching for Related Work	7 days	Mon 10/20/25	Sun 10/26/25
11	3.2	Techniques Table	5 days	Mon 10/27/25	Fri 10/31/25
12	4	**Phase 4 : Use Case Modeling**	**16 days**	**Sat 11/1/25**	**Sun 11/16/25**
13	4.1	Use Case Diagram	6 days	Sat 11/1/25	Thu 11/6/25
14	4.2	Writing Use Case Scenario	5 days	Fri 11/7/25	Tue 11/11/25
15	4.3	Activity Diagram	5 days	Wed 11/12/25	Sun 11/16/25
16	5	**Phase 5 : System Design Diagrams**	**23 days**	**Mon 11/17/25**	**Tue 12/9/25**
17	5.1	Class Diagram	5 days	Mon 11/17/25	Fri 11/21/25
18	5.2	ERD (Entity Relationship Diagram)	5 days	Sat 11/22/25	Wed 11/26/25
19	5.3	Sequence Diagram	6 days	Thu 11/27/25	Tue 12/2/25
20	5.4	Deployment Diagram	4 days	Wed 12/3/25	Sat 12/6/25
21	5.5	Block Diagram	3 days	Sun 12/7/25	Tue 12/9/25
22	6	**Phase 6 : IoT & Buisness Modeling**	**11 days**	**Wed 12/10/25**	**Sat 12/20/25**
23	6.1	Buisness Model	5 days	Wed 12/10/25	Sun 12/14/25
24	6.2	IoT Components Analysis	2 days	Mon 12/15/25	Tue 12/16/25
25	6.3	IoT Simulation	4 days	Wed 12/17/25	Sat 12/20/25
26	7	**AI Research & Evaluation**	**15 days**	**Tue 12/23/25**	**Tue 1/6/26**
27	7.1	Dataset Collection	5 days	Tue 12/23/25	Sat 12/27/25
28	7.2	Testing Existing AI Models	6 days	Sun 12/28/25	Fri 1/2/26
29	7.3	Accuracy Evaluation & Analysis	4 days	Sat 1/3/26	Tue 1/6/26
30	8	**Phase 8 : Risk & Finalization**	**4 days**	**Wed 1/7/26**	**Sat 1/10/26**
31	8.1	Risk Management	2 days	Wed 1/7/26	Thu 1/8/26
32	8.2	Final Review & Documentation Cleanup	2 days	Fri 1/9/26	Sat 1/10/26



	WBS	Task Name	Duration	Start	Finish
35	9	**Phase 9 : System Implementation**	**113 days**	**Tue 2/10/26**	**Tue 6/2/26**
36	9.1	Backend Development	29 days	Tue 2/10/26	Tue 3/10/26
37	9.2	AI Model Implementation	27 days	Wed 3/11/26	Mon 4/6/26
38	9.3	Frontend Web Application	26 days	Tue 4/7/26	Sat 5/2/26
39	9.4	Mobile Application Developpment	15 days	Sun 5/3/26	Sun 5/17/26
40	9.5	System Integration	11 days	Mon 5/18/26	Thu 5/28/26
41	9.6	Testing & Optimization	5 days	Fri 5/29/26	Tue 6/2/26
42	10	**Phase 10 : Deployment & Final Submission**	**10 days**	**Thu 6/11/26**	**Sat 6/20/26**
43	10.1	Deployment & Final Submission	10 days	Thu 6/11/26	Sat 6/20/26

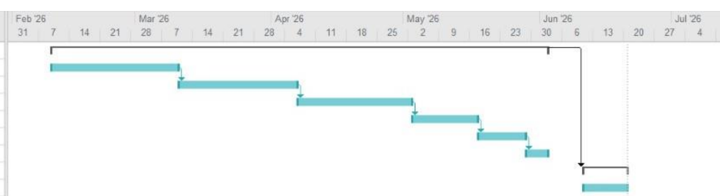


Figure 3.12 : Project Plan

3.5. Risk Management

This risk management table identifies the main technical, operational, and legal risks of the AI-based system and explains how each risk is handled. High-priority risks such as computational **cost**, overfitting, poor data quality, and false alarms are mainly addressed through minimisation strategies like optimized training, data validation, and model tuning. Contingency strategies are used for issues such as data availability, sensor failure, and network connectivity to ensure system continuity. Privacy and legal compliance is treated as a critical risk with a high impact, managed through avoidance measures such as anonymization and strict access control, ensuring the system remains reliable, accurate, and compliant.

Table 3.21 : Risk Management

Risk	Strategy	Type of strategy?	Priority	Probability	Impact (Effect)
High computational cost to train AI models	Use transfer learning, pre-trained models, and optimized training resources.	Minimisation	High	Medium	Serious
Overfitting of AI models	Use regularization, cross-validation, data augmentation, and early stopping.	Minimisation	High	Medium	Serious
Data availability	Use public datasets, synthetic data, and sensor-collected data.	Contingency	Medium	Medium	Serious
Poor data quality (noise, missing labels, bias)	Clean, validate, and normalize data before training.	Minimisation	High	High	Serious
AI model produces false alarms (false positives)	Tune thresholds, balance data, add rule-based checks and human verification.	Minimisation	High	High	Serious
Sensor hardware failure	Use reliable vendors and keep spare sensors available.	Contingency	Medium	Low	Moderate
Sensor installation or calibration errors	Perform testing, calibration, and regular maintenance.	Minimisation	Medium	Medium	Moderate
Network connectivity issues	Use local buffering and backup connections.	Contingency	Medium	Medium	Moderate
Privacy and legal compliance issues	Apply anonymization, access control, and follow regulations.	Avoidance	High	Low	Catastrophic
Integration issues between AI models and sensors	Use standard protocols and early integration testing.	Minimisation	Medium	Medium	Moderate

3.6. Technologies & Tools

Figma is a cloud-based design and prototyping tool used for user interface and user experience (UI/UX) design. It was chosen because it enables collaborative design, rapid prototyping, and easy sharing of interface layouts. Figma helps visualize the web dashboard and mobile application before development, ensuring usability, consistency, and clear communication between team members.

Visual Paradigm is a professional modeling and design tool used for software architecture and system modeling. It was selected for its strong support for UML diagrams, ER diagrams, and system design documentation in a collaborative environment. Visual Paradigm helps clearly define system structure, workflows, and interactions, improving system planning, documentation quality, and team coordination.

Google Colab is a cloud-based development environment that provides free access to GPU and TPU resources for machine learning tasks. It was chosen because it allows fast training, testing, and experimentation with AI models without requiring powerful local hardware. Google Colab simplifies collaboration, model prototyping, and experimentation, making it ideal for developing and validating deep learning models used in the surveillance system.

Python is a high-level programming language widely used in Artificial Intelligence and backend development. It was chosen because of its strong AI ecosystem, extensive libraries, and ease of integration with cameras, sensors, and backend services. Its simplicity reduces development time while maintaining flexibility and scalability.

PyTorch is an open-source deep learning framework used for building and running neural networks. It was selected due to its flexibility, dynamic computation graph, and efficient GPU support, which make it suitable for developing and fine-tuning real-time AI models used in surveillance and behavior analysis.

OpenCV is an open-source computer vision library designed for real-time image and video processing. It was chosen for its ability to efficiently handle camera streams, perform frame processing, and support motion and image analysis, making it essential for live surveillance video handling.

FastAPI is a modern backend web framework used for building high-performance APIs. It was chosen for its asynchronous processing capabilities, high speed, and built-in support for secure API development, making it suitable for managing AI inference, sensor data processing, and real-time alert delivery.

Arduino IDE is a development environment used for programming microcontrollers and embedded systems. It was selected for its simplicity, hardware compatibility, and reliable sensor integration, enabling efficient collection and transmission of environmental sensor data to trigger AI-based actions.

Flutter is a cross-platform mobile application development framework. It was chosen because it allows building high-performance mobile applications using a single codebase, supports real-time updates, and provides smooth user interfaces for monitoring and receiving alerts on mobile devices.

React is a JavaScript-based frontend library used for building interactive user interfaces. It was selected for its component-based architecture, efficient state management, and real-time UI update capabilities, making it ideal for developing a responsive and scalable surveillance dashboard.

Git is a version control system, and GitHub is a cloud-based platform for code hosting and collaboration. They were chosen to manage source code efficiently, support teamwork, track changes, and maintain project stability through version control and collaborative development tools.

Tinkercad is an online simulation and design tool used for electronics and embedded systems prototyping. It was chosen because it allows virtual simulation of Arduino circuits and sensors before physical implementation. Tinkercad helps test sensor connections, logic behavior, and basic programs safely and quickly, reducing hardware errors, cost, and development time during the early stages of the project.

CHAPTER 4

4. EXPERIMENTAL RESULTS

4.1. Simulation Results

4.1.1. Gas and Smoke Sensor Experiment (MQ-2)

In this experiment, a gas and smoke sensor (MQ-2) is used to detect the presence of gas or smoke leakage in the environment.

The sensor reads analog values representing gas concentration and compares them with a predefined threshold.

- The gas sensor value is read from an analog pin.
- If the gas value exceeds the threshold value, the system identifies a dangerous condition.
- Red LED turns ON to indicate danger.
- Green LED turns OFF to indicate unsafe conditions.
- A buzzer is activated as an alarm.
- If the gas value is below the threshold, the environment is considered safe.
- Green LED turns ON and the buzzer remains OFF.

The sensor readings and system status are displayed on the Serial Monitor and updated every second.

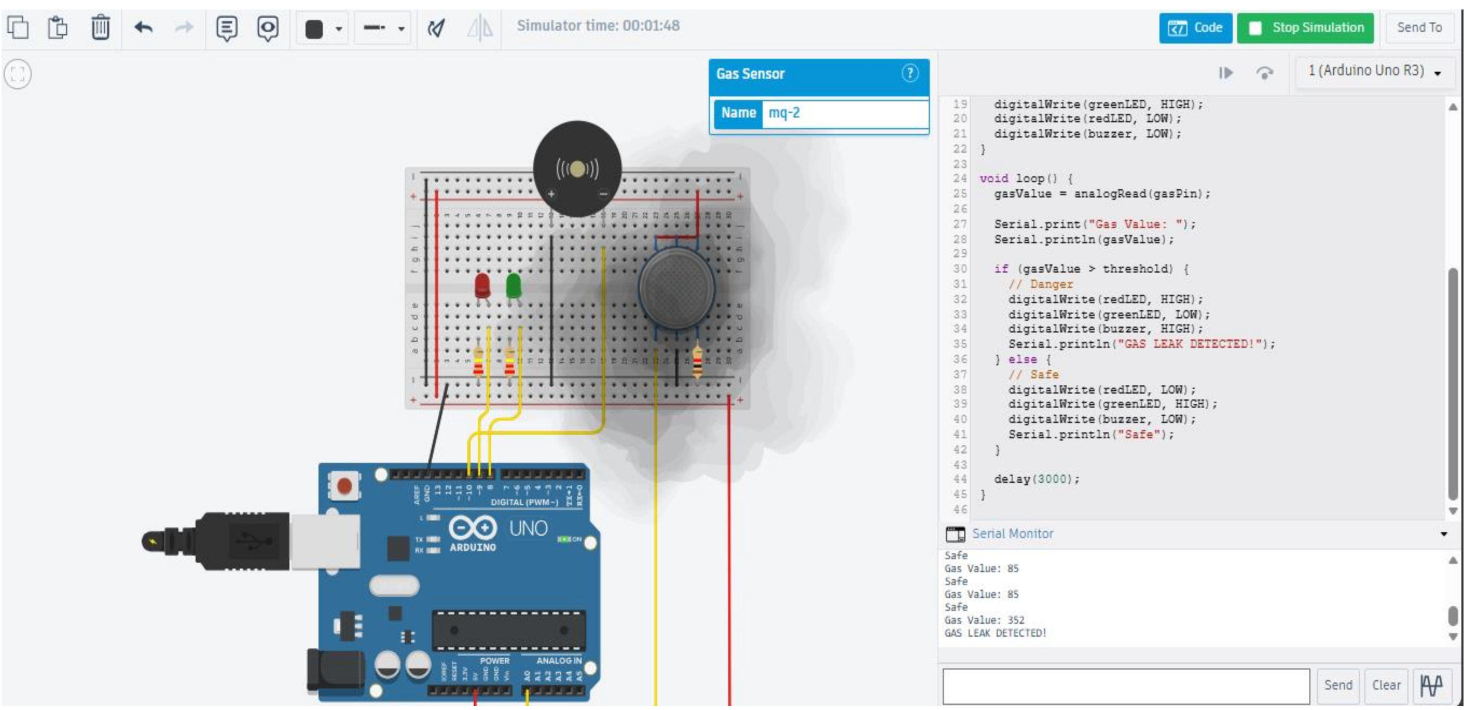


Figure 4.1 : Gas and Smoke Sensor (MQ-2)

4.1.2. Motion Sensor Experiment (PIR Sensor)

This experiment uses a PIR motion sensor to detect human movement within a monitored area.

- The PIR sensor provides a digital signal indicating motion detection.
- When motion is detected, the sensor output becomes HIGH.

- An LED turns ON to indicate detected movement.
- A message is displayed on the Serial Monitor indicating motion detection.
- When no motion is detected, the LED remains OFF.

The system continuously monitors motion and updates the status every x milliseconds.

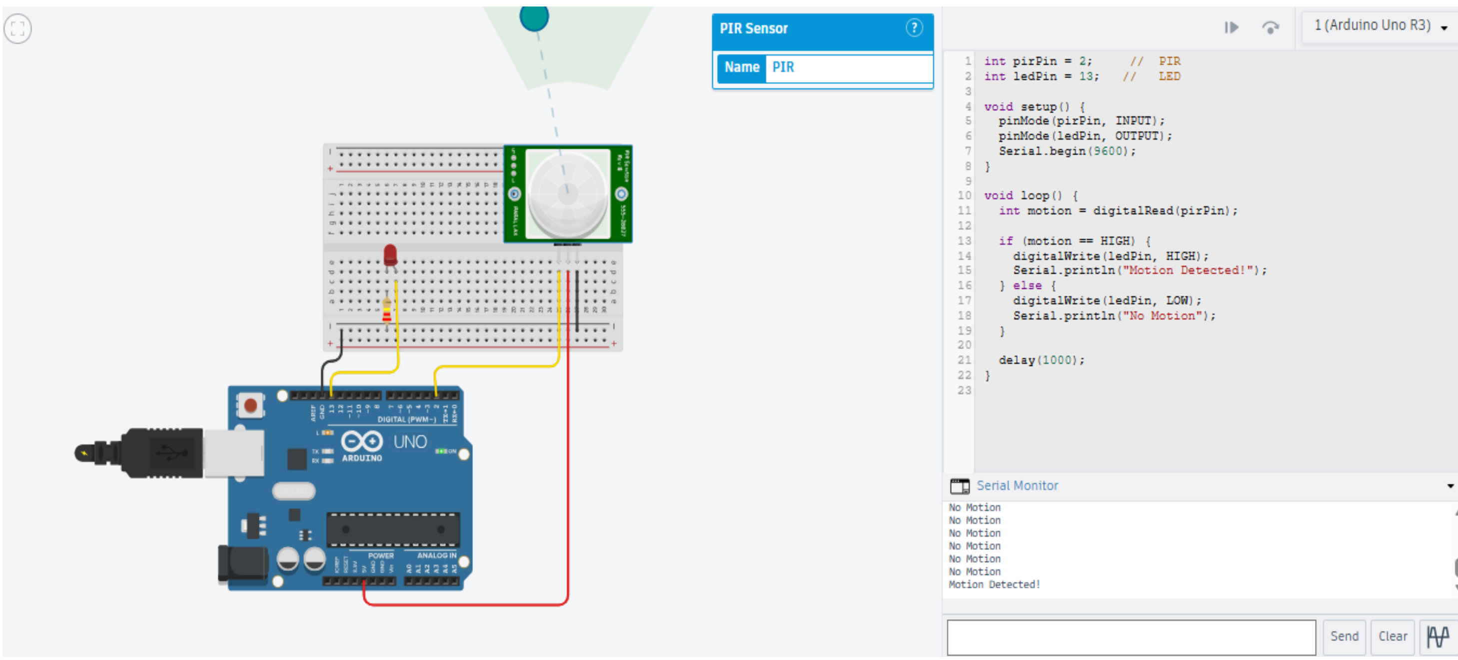


Figure 4.2 : Motion Sensor (PIR)

4.1.3. Temperature Sensor Experiment

In this experiment, a temperature sensor is used to measure ambient temperature.

- The sensor provides an analog output that is converted into voltage.
- The voltage is then converted into temperature values in degrees Celsius.

- The temperature value is displayed on the Serial Monitor.
- If the temperature exceeds 40°C, an alert condition is triggered.
- An LED turns ON to indicate high temperature.
- If the temperature is below the threshold, the LED remains OFF.

The temperature readings are updated every second to ensure continuous monitoring.

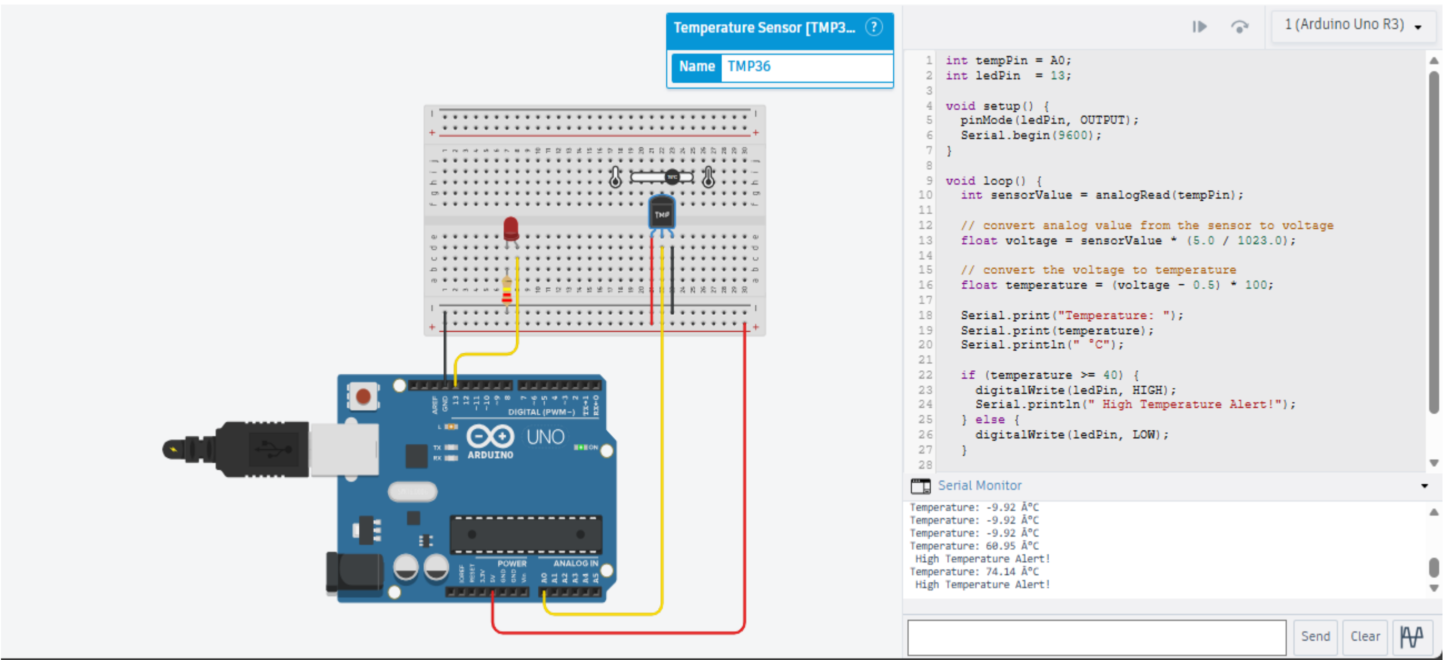


Figure 4.3 : Temperature Sensor

4.1.4. Sound and Vibration Sensor Experiment (KY-038)

This experiment uses a sound and vibration sensor (KY-038) to detect sound intensity in the environment.

- The sensor reads analog values representing sound levels.

- The sound level is compared to a predefined threshold.
- If the sound level exceeds the threshold, the system detects sound activity.
- An LED turns ON to indicate detected sound or vibration.
- A message is displayed on the Serial Monitor.
- If the sound level is below the threshold, the LED remains OFF.

The system continuously monitors sound levels and updates the readings every second.

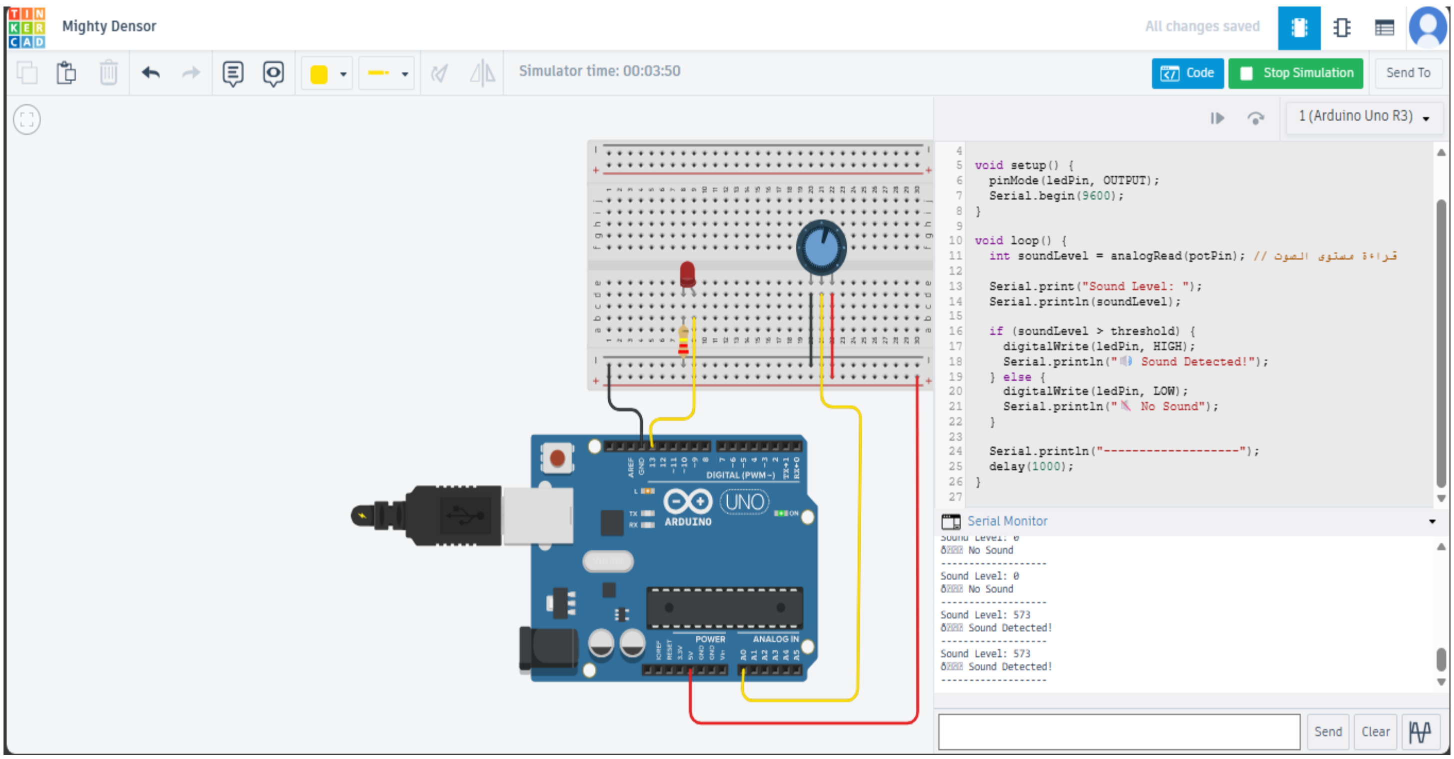


Figure 4.4 : Sound and Vibration Sensor (KY-038)

4.1.5. All System Simulation.

In this simulation, an integrated IoT system is designed to monitor environmental and security conditions using multiple sensors, including gas, motion, temperature, and sound sensors.
The system reads all sensor data continuously and provides real-time alerts using LEDs, a buzzer, and Serial Monitor messages.

- Gas and Smoke Monitoring:
  The MQ-2 sensor detects gas concentration and compares it with a threshold.
  If a gas leak is detected, the red LED and buzzer turn ON, and the green LED turns OFF.
  When conditions are safe, the green LED remains ON.

- Motion Detection:
  The PIR sensor detects movement and activates a motion LED when motion is present.
  Motion status is displayed on the Serial Monitor.

- Temperature Monitoring:
  The temperature sensor measures ambient temperature and converts it to degrees Celsius.
  If the temperature exceeds 40°C, a temperature LED turns ON indicating a high-temperature alert.

- Sound Monitoring:
  The sound sensor detects sound intensity.
  When the sound level exceeds the threshold, a sound LED turns ON.

All sensors operate together in a single loop, providing continuous monitoring and real-time feedback.
This simulation demonstrates a simple and effective IoT-based smart monitoring system.

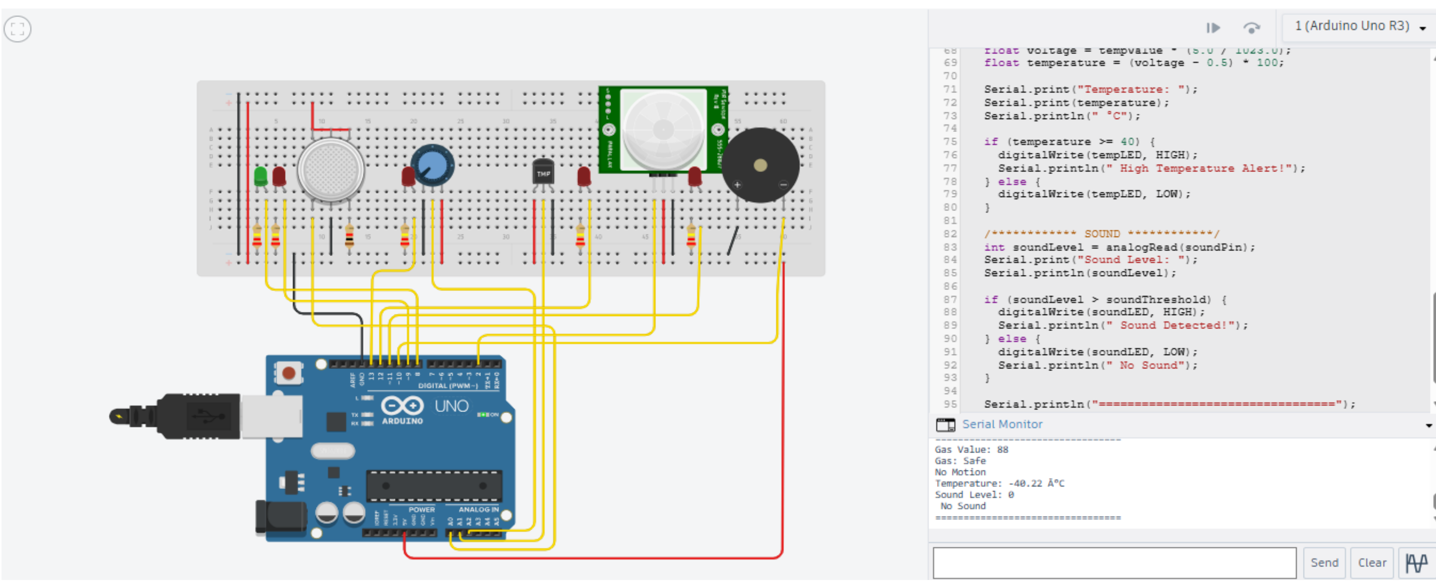


Figure 4.5: All System Simulation

4.2. AI Results

At the initial stage of this research, we aimed to improve the accuracy and reliability of the proposed Intelligent Surveillance & Safety System by evaluating multiple machine learning and deep learning models across different security-related visual scenarios. The primary objective was to develop a robust system capable of detecting and recognizing abnormal and suspicious activities under challenging surveillance conditions, including low resolution, occlusion, illumination changes, and crowded environments.

To achieve this, we evaluated a range of classification approaches, including both classical machine learning models and deep learning architectures. For image-based tasks such as weapon detection, fire and smoke recognition, and restricted-area violations using facial analysis, we employed Support Vector Machines (SVM) with HOG features, Convolutional Neural Networks (CNNs), and transfer learning models such as ResNet50. For video-based activity recognition, we assessed C3D, Two-Stream CNN, and I3D architectures, which integrate spatial and temporal feature learning to effectively capture dynamic patterns in surveillance footage.

Each model was trained and evaluated using labeled surveillance images and videos corresponding to different security scenarios. The performance of these models was assessed using standard evaluation metrics including precision, recall, F1-score, and accuracy, with particular emphasis on recall and F1-score for critical classes such as weapons, fire, smoke, and restricted-area violations, where false negatives can pose serious safety risks.

Initial experiments revealed noticeable differences in model performance depending on the task complexity and visual characteristics of each class even with deep learning approaches, some challenges remained in distinguishing visually similar classes, such as fire versus smoke or authorized versus unauthorized individuals, highlighting the difficulty of real-world surveillance environments.

The following tables summarize the performance of the evaluated video-based classifiers on our surveillance activity dataset:

Table 4.1 : Activity Classification Results

Classifier	Class	Precision	Recall	F1-Score	Accuracy
C3D	Abuse	68 %	66 %	67 %	71 %
	Arson	70 %	68 %	69 %	
	ViolentActs	69 %	66 %	67 %	
	Theft/Robbery	66 %	64 %	65 %	
	Explosion	70 %	68 %	69 %	
	Vandalism	68 %	66 %	67 %	
	CameraTampering	72 %	69 %	70 %	
	Normal	72 %	69 %	70 %	
	Atypical	65 %	62 %	63 %	
Two-Stream CNN	Abuse	70 %	67 %	68 %	73 %
	Arson	72 %	70 %	71 %	
	ViolentActs	71 %	68 %	69 %	
	Theft/Robbery	69 %	66 %	67 %	
	Explosion	72 %	70 %	71 %	
	Vandalism	71 %	68 %	69 %	
	CameraTampering	74 %	71 %	72 %	
	Normal	74 %	71 %	72 %	
	Atypical	67 %	64 %	66 %	
I3D	Abuse	73 %	70 %	71 %	76 %
	Arson	74 %	71 %	72 %	
	ViolentActs	73 %	70 %	71 %	
	Theft/Robbery	71 %	68 %	69 %	
	Explosion	74 %	71 %	72 %	
	Vandalism	73 %	70 %	71 %	
	CameraTampering	74 %	71 %	72 %	
	Normal	74 %	71 %	72 %	
	Atypical	69 %	66 %	67 %	

Table 4.2 : Restricted Area Violation Results

Model	Class	Precision	Recall	F1-Score	Accuracy

SVM	Violation	55 %	50 %	52 %	61 %
	Non-Violation	70 %	73 %	71 %	
CNN	Violation	58 %	54 %	56 %	66 %
	Non-Violation	72 %	74 %	73 %	
ResNet50	Violation	62 %	60 %	61 %	71 %
	Non-Violation	71 %	72 %	71 %	

Table 4.3 : Weapon Detection Results

Model	Class	Precision	Recall	F1-Score	Accuracy
SVM	Weapon	50 %	45 %	47 %	63 %
	Non-Weapon	65 %	68 %	66 %	
CNN	Weapon	55 %	52 %	53 %	67 %
	Non-Weapon	68 %	70 %	69 %	
ResNet50	Weapon	58 %	55 %	56 %	69 %
	Non-Weapon	69 %	70 %	69 %	

Table 4.4 : Fire and Smoke Detection Results

Model	Class	Precision	Recall	F1-Score	Accuracy
SVM	Normal	65%	68%	66%	64%
	Fire	53%	47%	50%	
	Smoke	48%	44%	46%	
CNN	Normal	72%	75%	73%	69%
	Fire	60%	55%	57%	
	Smoke	58%	53%	55%	
ResNet50	Normal	77%	80%	78%	73%
	Fire	65%	60%	62%	
	Smoke	63%	58%	60%	

By comparing the results of different models, we identified patterns, strengths, and weaknesses for each approach. This multi-algorithm evaluation was a key step in optimizing our final surveillance system and guiding the development of our models.

4.3. Design Mockup

This design mockup presents the complete user interface flow of the smart surveillance system, covering essential screens such as user authentication, live camera monitoring, event detection alerts, incident history, analytics dashboard, and system settings. Each screen is designed to deliver a smooth and user-friendly experience, focusing on real-time monitoring, intelligent threat detection, and quick response actions through clear navigation and a modern, professional interface.

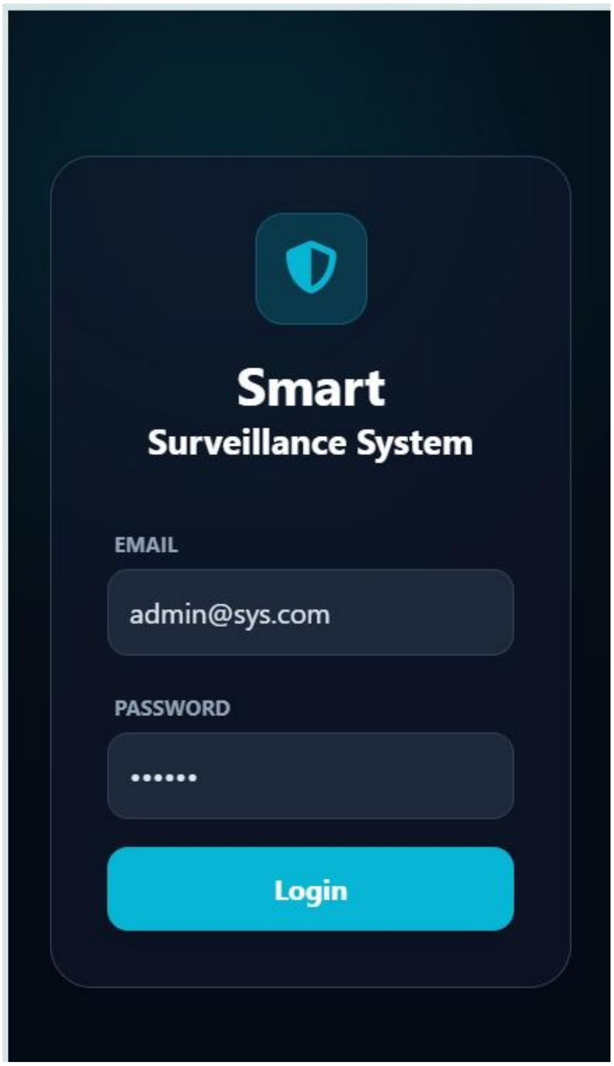


Figure 4.6 : Admin Login UI

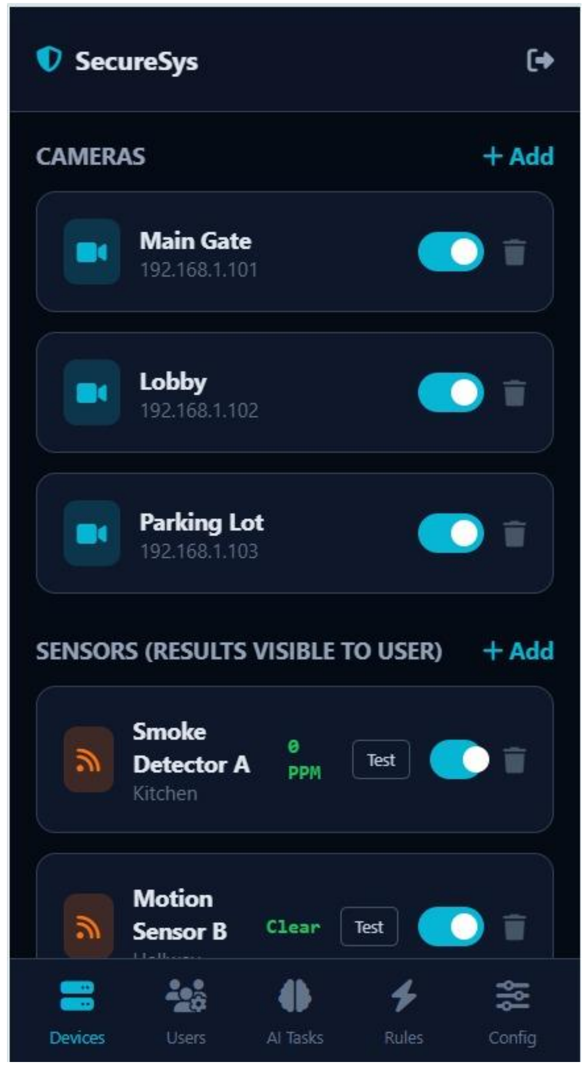
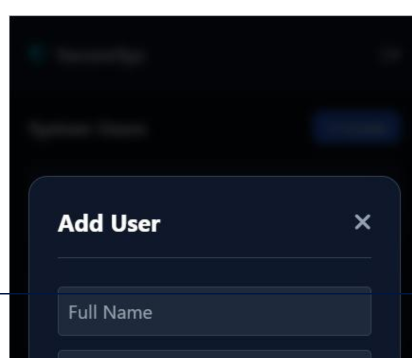


Figure 4.7 : Managing Cameras & Sensors UI



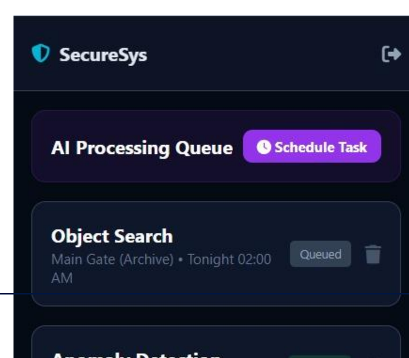
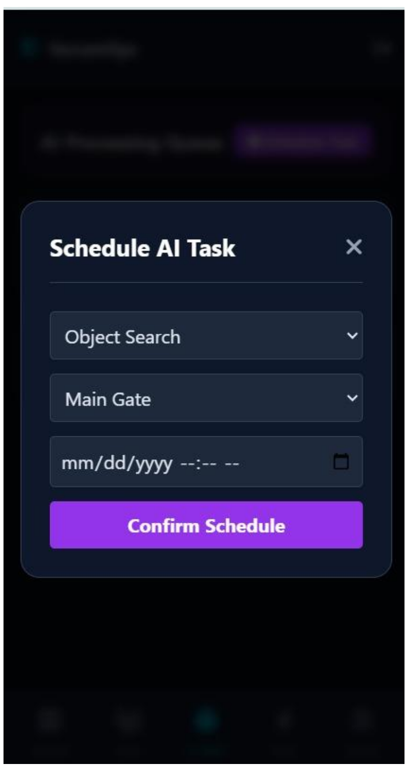
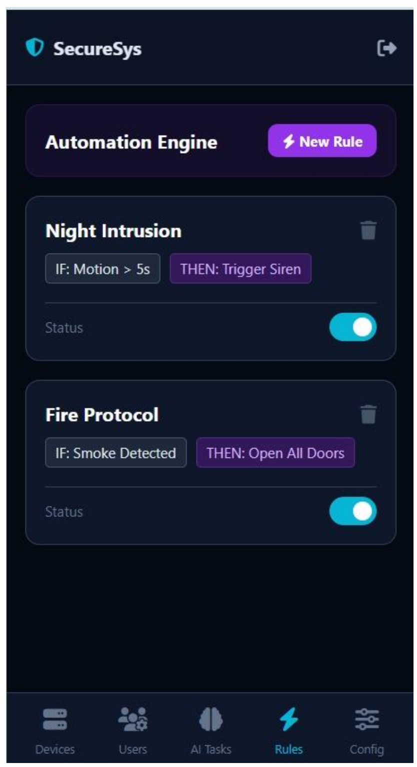


Figure 4.10 : Manage Users UI

Figure 4.8 : Add Users UI

Figure 4.9 : Scheduling Tasks Management UI





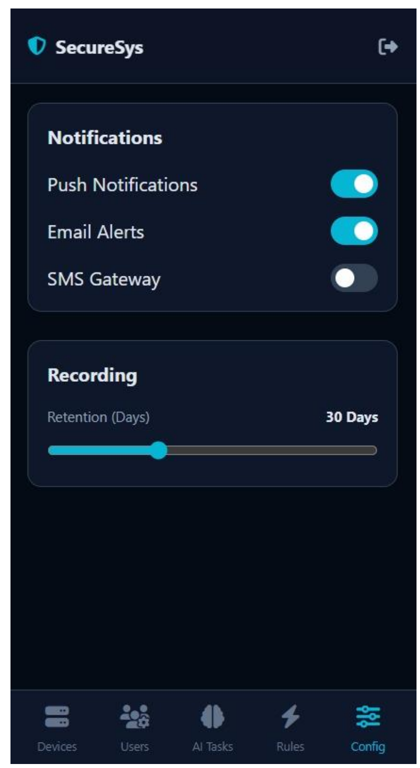
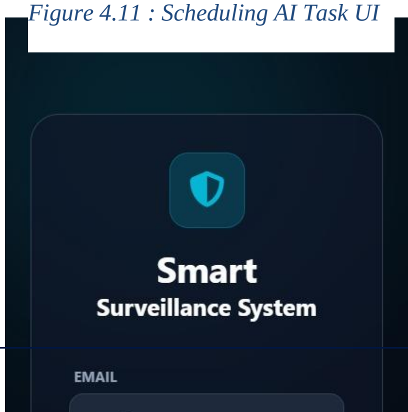
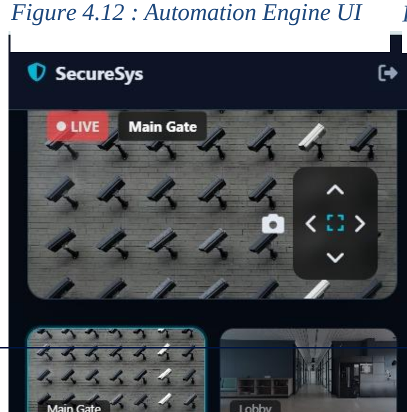


Figure 4.11 : Scheduling AI Task UI

Figure 4.12 : Automation Engine UI

Figure 4.13 : Manage Recording & Notifications UI





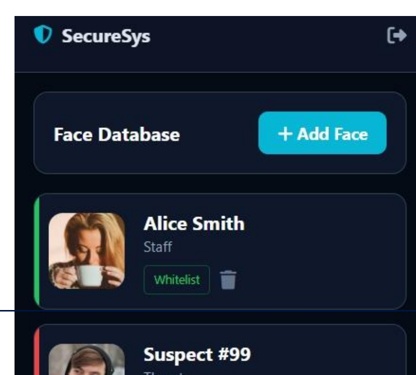


Figure 4.15 : User Login UI

Figure 4.14 : Live Cameras and Sensor Result UI

Figure 4.16 : Face Management UI

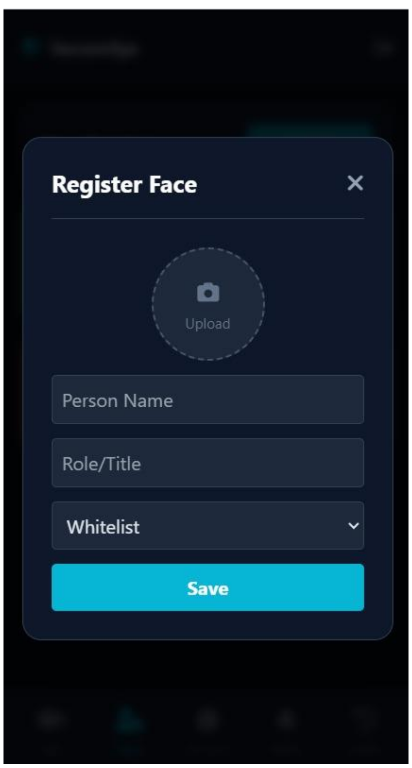


Figure 4.17 : Whitelist & Blacklist Management UI

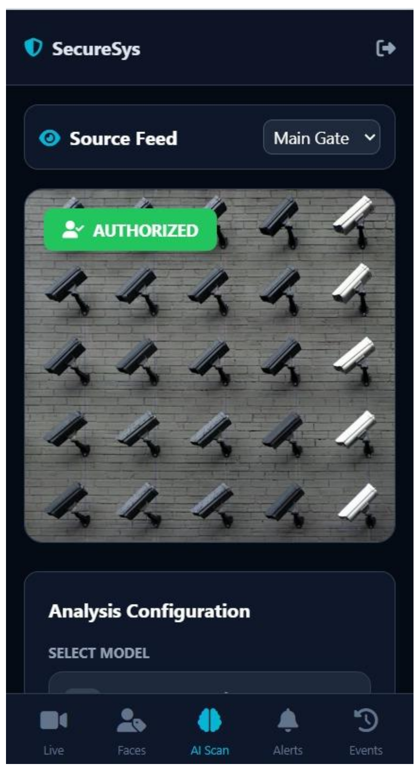


Figure 4.18 : AI Scan UI

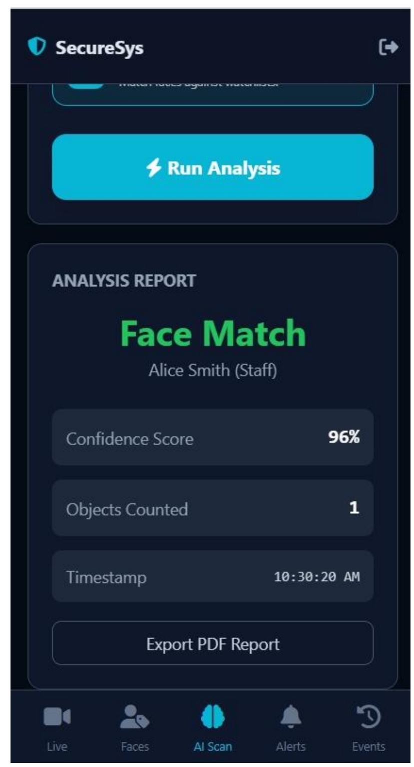
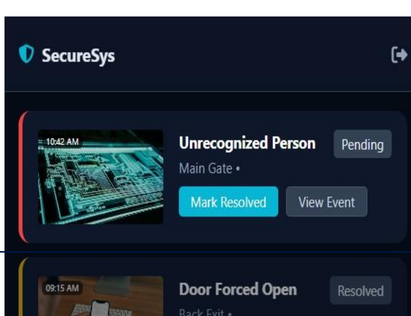
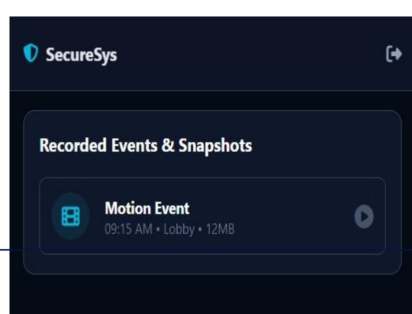


Figure 4.19 : AI Scan Result UI





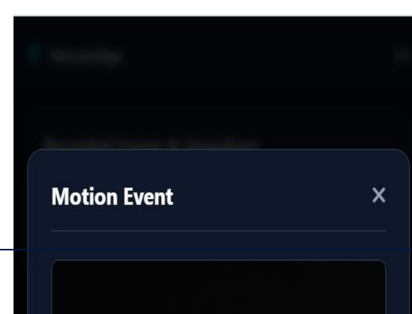
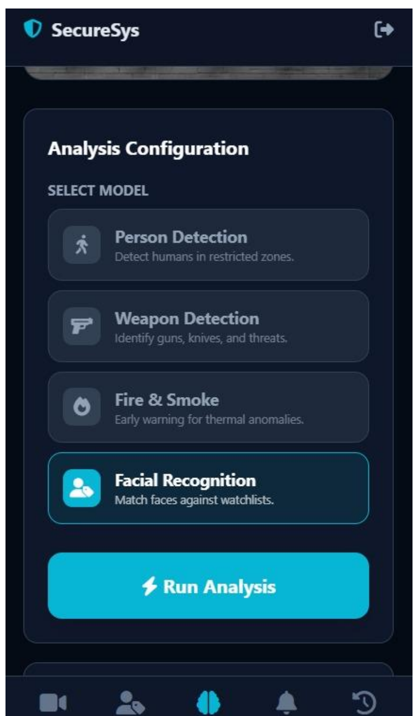


Figure 4.20 : Manage Alerts UI

Figure 4.21: Recorded Events & Snapshot UI

Figure 4.22 : Watch & Download Events UI



CONCLUSION

This project presents an intelligent and integrated Smart Surveillance and Safety System designed to enhance security and risk management through the use of Artificial Intelligence, Computer Vision, and IoT technologies. The proposed system overcomes the limitations of traditional surveillance methods by enabling automated threat detection, real-time monitoring, and context-aware decision making.

The system integrates live camera streams with environmental sensors to detect a wide range of abnormal and suspicious activities, including fire and smoke incidents, gas leaks, weapon detection, unauthorized access, and violent behavior. A key strength of the proposed framework is its event-driven architecture, where sensor data intelligently triggers AI models only when potential risks are detected. This approach significantly reduces unnecessary processing, optimizes system resources, and improves response time.

Furthermore, the system provides an advanced centralized dashboard that allows authorized users to monitor live feeds, receive real-time alerts, manage users and devices, configure automation rules, and review recorded events. The use of role-based access control and encrypted communication ensures system security and data privacy.

By combining sensor fusion, selective AI activation, and intelligent event management, the proposed system delivers a scalable, efficient, and reliable surveillance solution suitable for residential, commercial, industrial, healthcare, and public environments. This project successfully bridges the gap between traditional monitoring systems and modern smart security technologies, contributing to a safer and more proactive surveillance future.

In the next phase of this project, practical implementation will begin starting from the second academic term, where the system will move from a prototype design to a real-world deployment. This stage will include selecting and purchasing suitable surveillance cameras and environmental sensors, such as smoke, gas, motion, temperature, and sound sensors, to build a fully operational hardware setup. Proper installation and calibration of these devices will be performed to ensure accurate data collection and reliable system performance.

Future development will focus on expanding the system’s capabilities by supporting additional AI detection modules, such as vehicle recognition, license plate detection, and advanced crowd behavior analysis. This will enhance system adaptability for large-scale environments such as airports, smart cities, and transportation hubs.

The system can also be improved by introducing edge computing, allowing AI processing to run directly on smart cameras or embedded devices. This will reduce network dependency, decrease latency, and improve real-time responsiveness, especially in critical situations.

Another important enhancement will be the integration of predictive analytics, where historical sensor and video data will be analyzed to forecast potential risks before they occur. This proactive approach can help prevent incidents instead of only reacting to them.

Moreover, future versions of the platform can include multi-site management, enabling organizations to monitor and control multiple locations from a single dashboard. Additional features such as voice-controlled commands, offline monitoring with automatic synchronization, and detailed energy consumption analytics can also be implemented to improve usability and system efficiency.

Finally, further research can explore cloud scalability, AI model optimization, and privacy-preserving techniques such as face anonymization to ensure ethical system deployment. These enhancements will make the surveillance system more intelligent, secure, and suitable for real-world large-scale applications.

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